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Bioinformatics Algorithms

Molecular biology - a quick (and not so accurate) introduction

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Bioinformatics

=

Bio(logy)

+

Informatics

- Important: **information science** $\neq$ **computer sciene** !
- Bioinformatics *includes* notions of computer science, but also chemistry, physics, information science/engineering, mathematics and statistics ...
- Information science is a field primarily concerned with the
  - analysis (encoding and decoding),
  - classification,
  - manipulation,
  - storage,
  - retrieval,
  - transmission,

  of information! (Adapted from Wikipedia)
  Here: biological information stored in living cells!
- Used for "*in silico*" analyses of biological data (from "*in vivo*" or "*in vitro*" sources)

Possible definitions of bioinformatics

Oxford English Dictionary:

> *"The science of information and information flow in biological systems, especially of the use of computational methods in genetics and genomics."*

Fredj Tekaia (Institut Pasteur, Paris):

> *"The mathematical, statistical and computing methods that aim to solve biological problems using DNA and amino acid sequences and related information."*

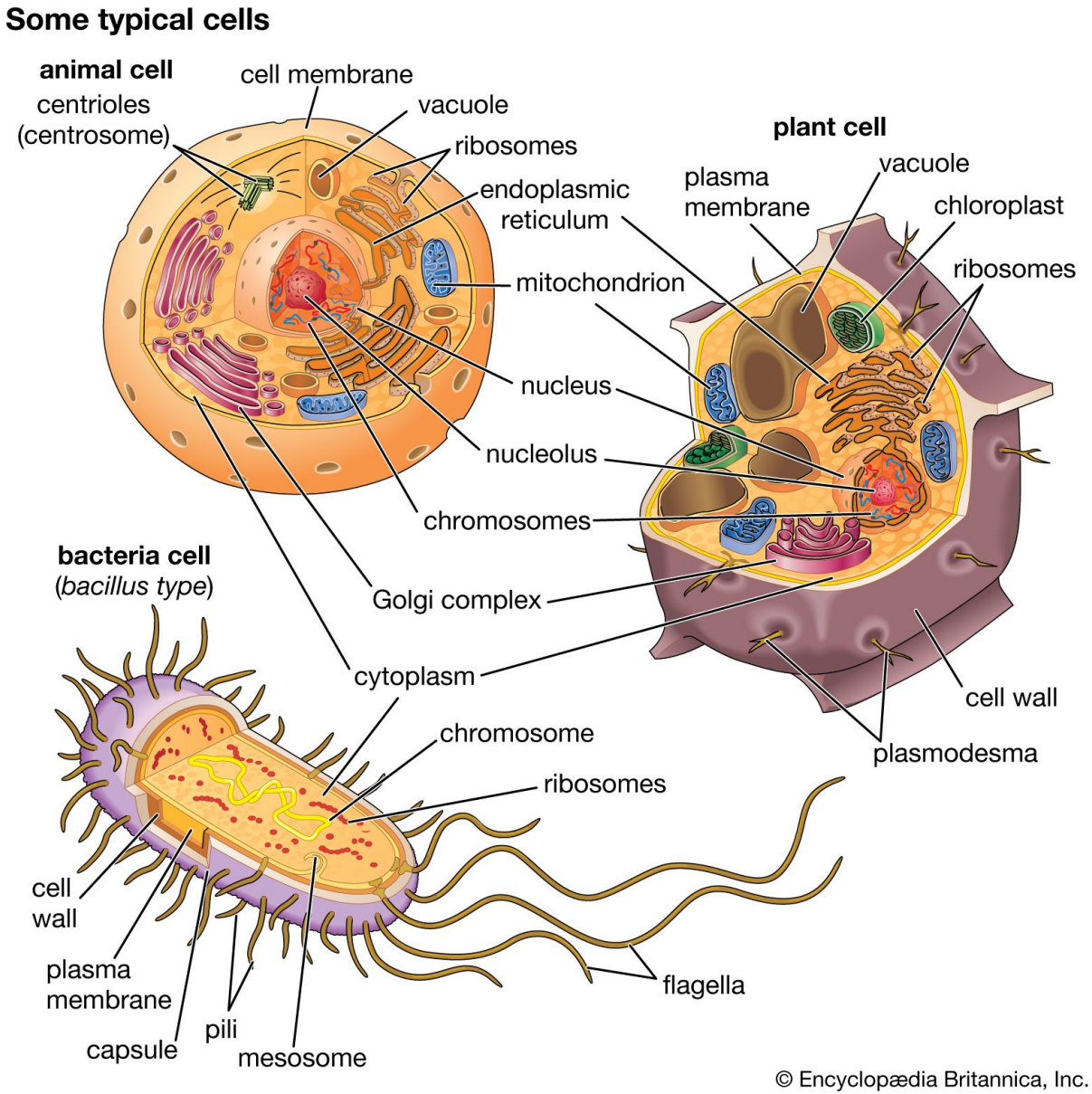
Wikipedia:

> *"... an interdisciplinary field that develops methods and software tools for understanding biological data, in particular when the data sets are large and complex. As an interdisciplinary field of science, bioinformatics combines biology, chemistry, physics, computer science, information engineering, mathematics and statistics to analyze and interpret the biological data."*

Bioinformatics **is not** biologically-inspired computation, e.g., genetic algorithms and neural networks, but it can *use* such methods to solve biological problems!

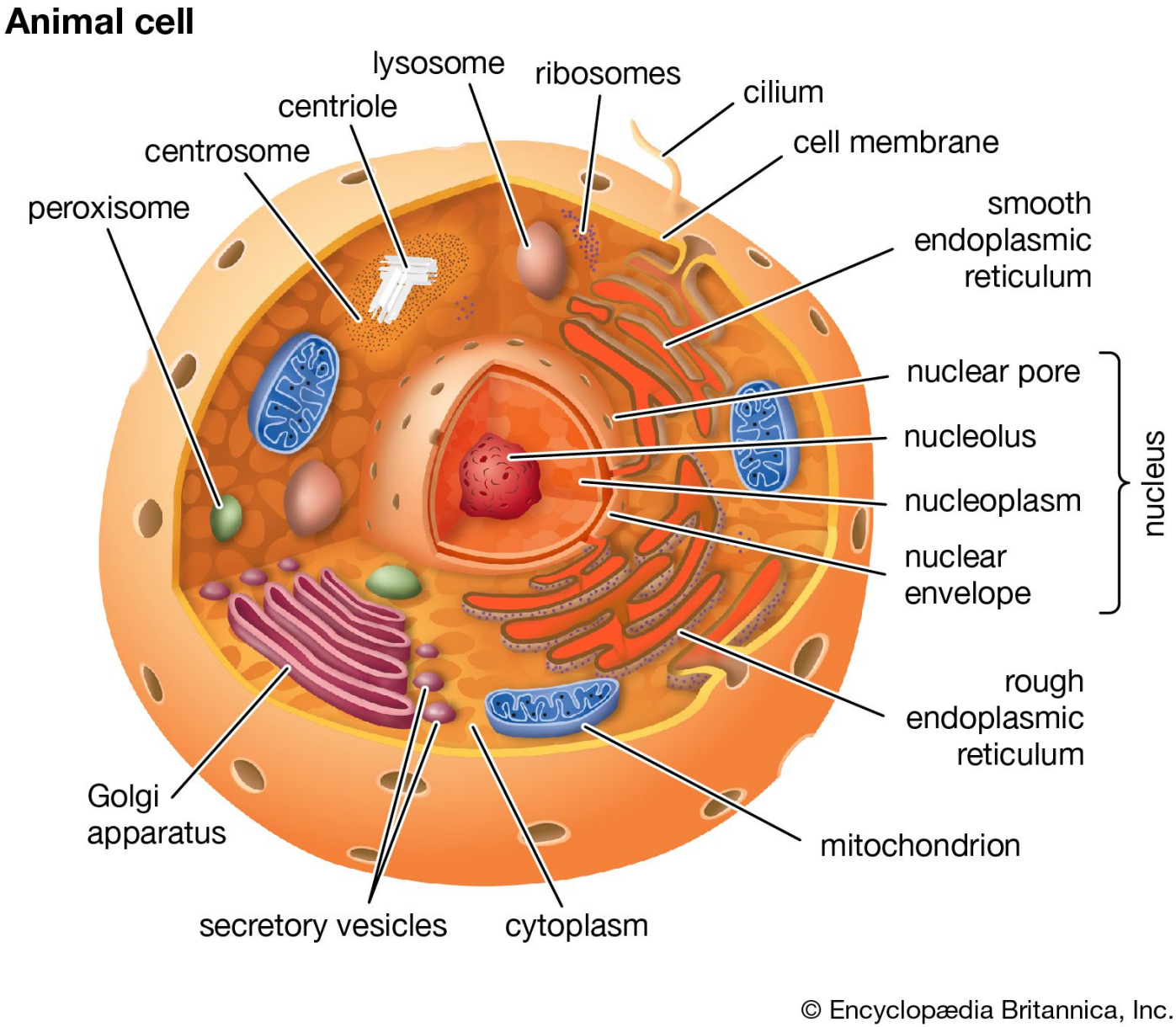
"Computational biology": application of computing to biology (broad definition)

Animal, plant and bacteria cells



(from: https://www.britannica.com/science/cell-biology)

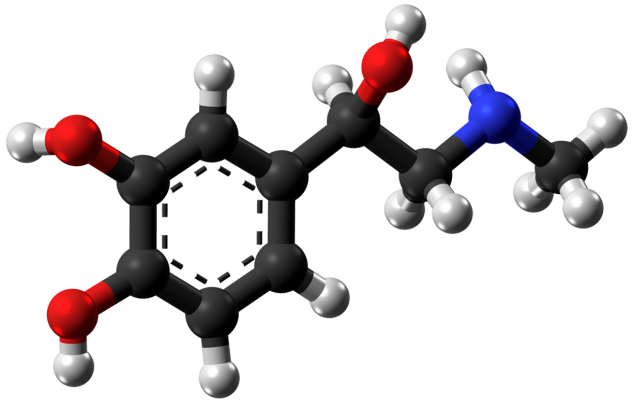
Animal cells



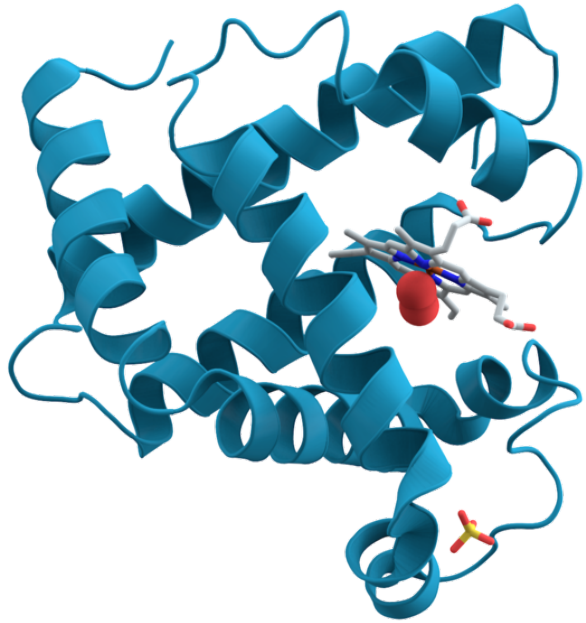
(from: https://www.britannica.com/science/cell-biology)

Molecules

- Small molecules
  - Sources of energy; “signaling molecules” for signal transmission; building blocks of macromolcules
    - ★ Water, fatty acids, sugars, nucleotides, amino acids, ...
- Proteins
  - Building blocks of the cell (and the organelles); functional molecules for biological processes (e.g., by catalyzing chemical reactions); signal transduction
    - ★ Structural proteins; enzymes; “receptors” on cell membranes, “transcription factors” which regulate gene expression
  - Often collaborate in “protein complexes” or have otherwise physical “protein–protein interactions”



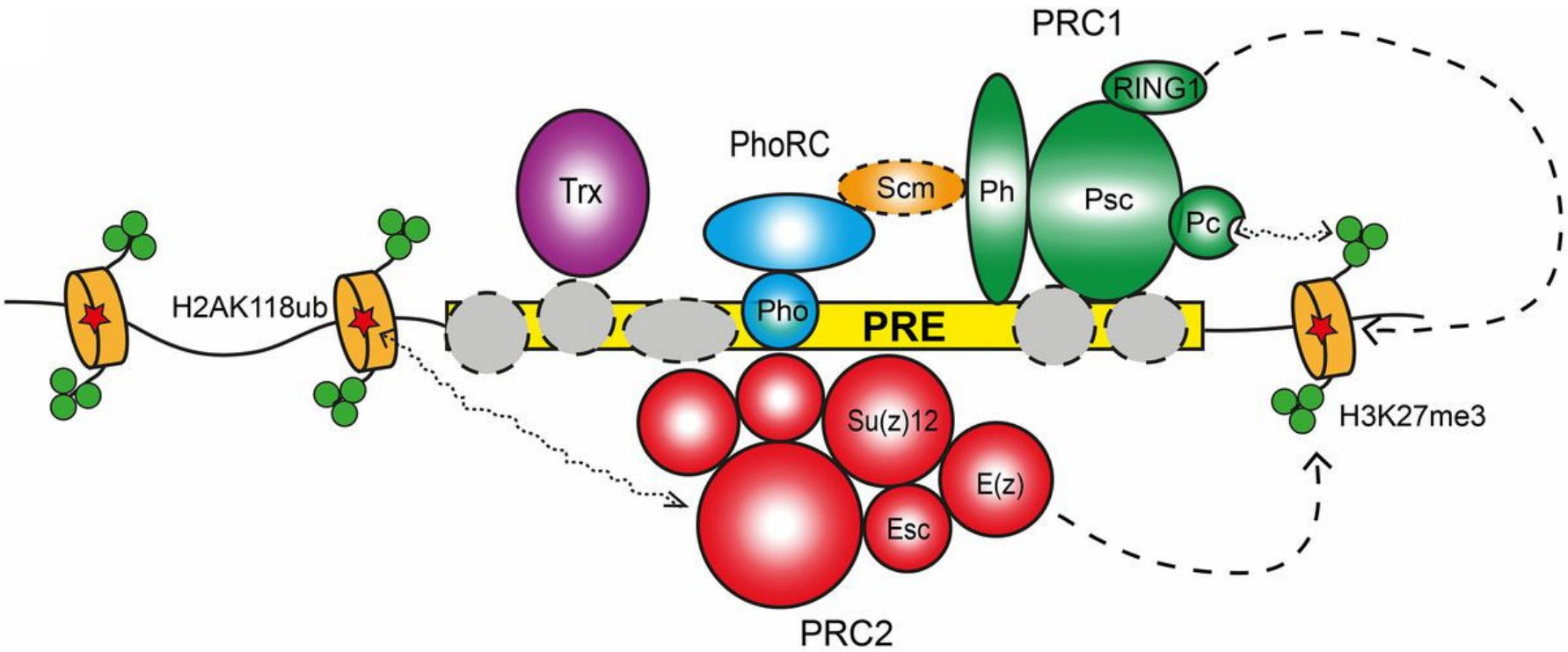
(adrenaline; from: Wikimedia Commons, author: Jynto)



(myoglobin; from: Wikipedia, author: Splette)

Molecules

Example for protein complexes

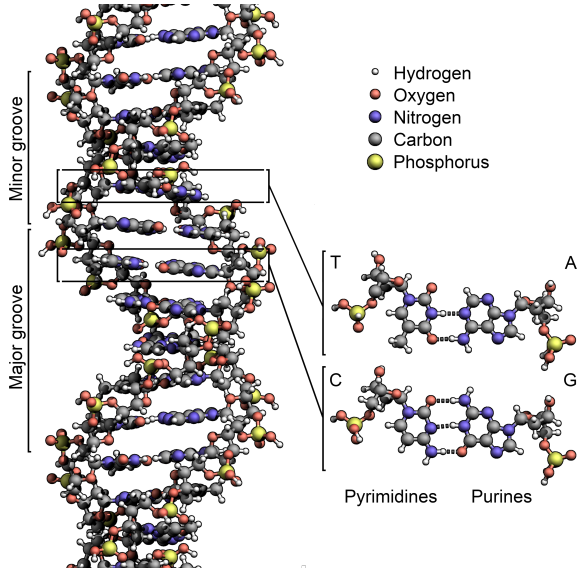


Polycomb complexes PRC1 and PRC2, and the PhoRC complex, interacting with a certain functional element in the genome, called a Polycomb response element (PRE)

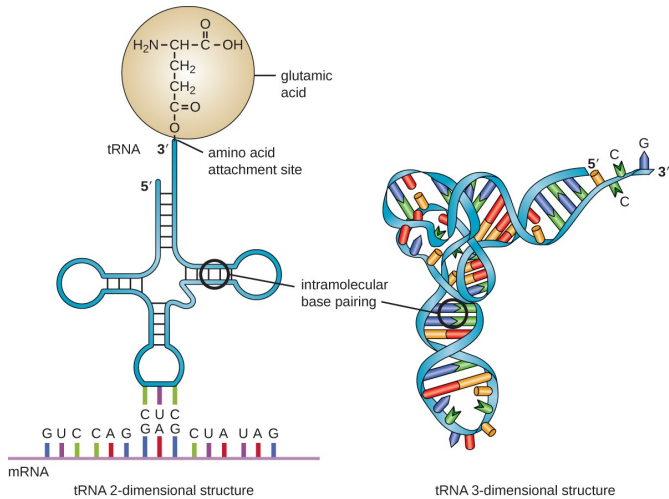
(from: Schwartz and Cavalli, 2017)

Molecules

- DNA (deoxyribonucleic acid)
  - Storage and reproduction of information
    - ★ Contains genes, regulatory elements (e.g., “transcription factor binding elements”) and “junk DNA” (regions without known function)
    - ★ Double-stranded (“double helix”)
- RNA (ribonucleic acid)
  - Key role in transformation of genetic information to function
    - ★ E.g., helps in constructing proteins (amino acid polymers) from genomic information stored in DNA
    - ★ Single-stranded



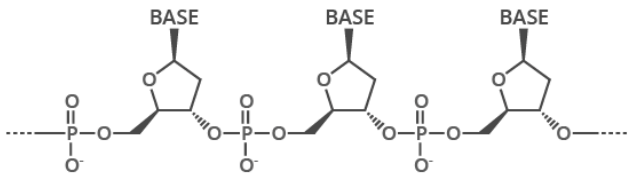
(from: Wikipedia, authors: Zephyris, JeffyP)



(tRNA; from: `https://courses.lumenlearning.com`)

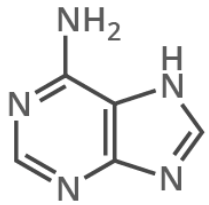
THE CHEMICAL STRUCTURE OF DNA

THE SUGAR PHOSPHATE 'BACKBONE'

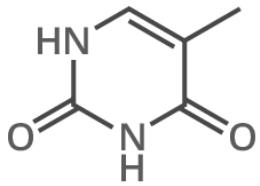


DNA is a polymer made up of units called nucleotides. The nucleotides are made of three different components: a sugar group, a phosphate group, and a base. There are four different bases: adenine, thymine, guanine and cytosine.

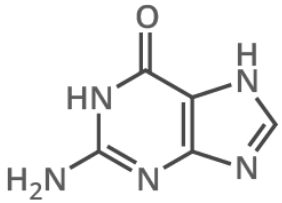
A ADENINE



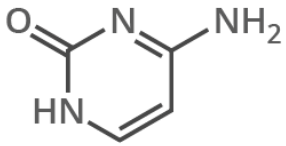
T THYMINE



G GUANINE



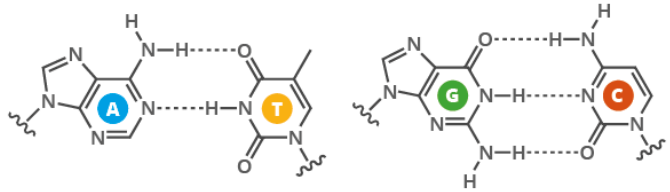
C CYTOSINE





WHAT HOLDS DNA STRANDS TOGETHER?

DNA strands are held together by hydrogen bonds between bases on adjacent strands. Adenine (A) always pairs with thymine (T), while guanine (G) always pairs with cytosine (C). Adenine pairs with uracil (U) in RNA.



FROM DNA TO PROTEINS

The bases on a single strand of DNA act as a code. The letters form three letter codons, which code for amino acids - the building blocks of proteins.



An enzyme, RNA polymerase, transcribes DNA into mRNA (messenger ribonucleic acid). It splits apart the two strands that form the double helix, then reads a strand and copies the sequence of nucleotides. The only difference between the RNA and the original DNA is that in the place of thymine (T), another base with a similar structure is used: uracil (U).

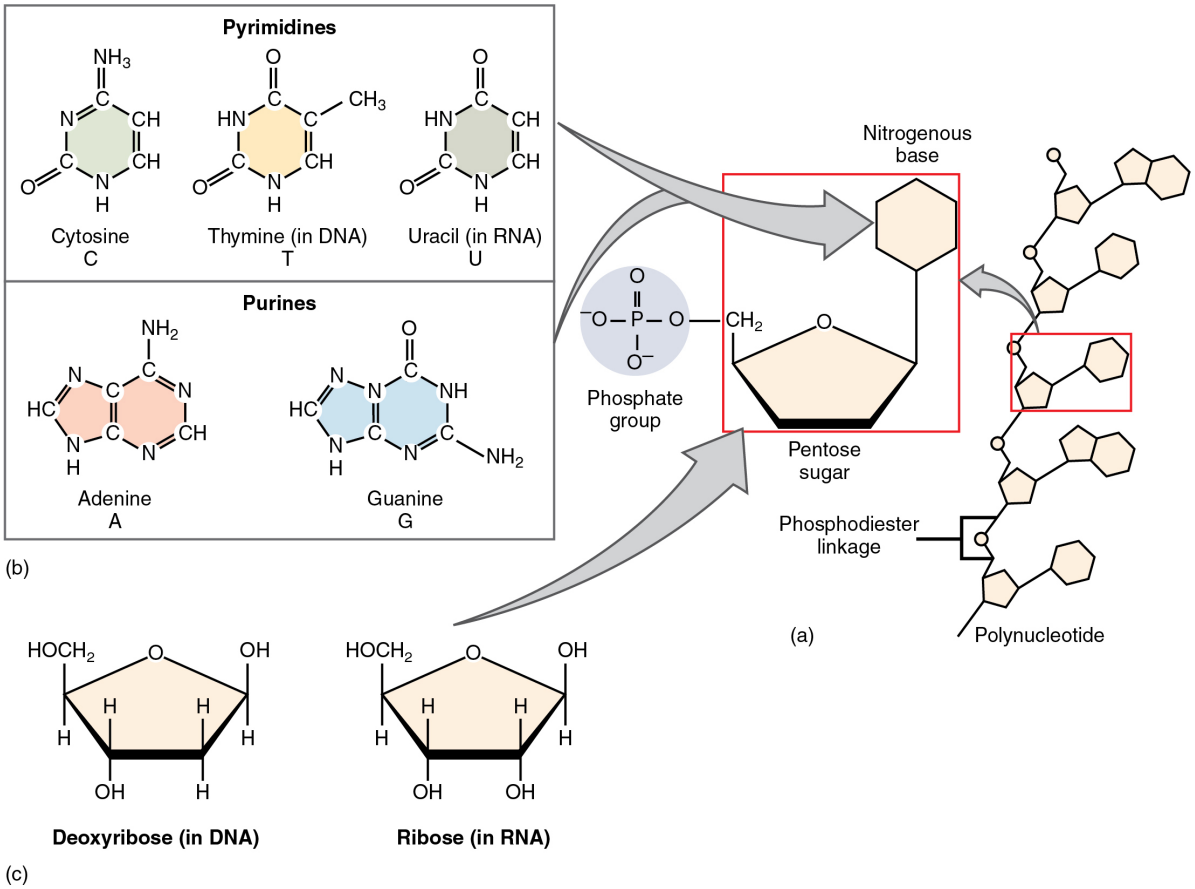


In multicellular organisms, the mRNA carries genetic code out of the cell nucleus, to the cytoplasm. Here, protein synthesis takes place. 'Translation' is the process of turning the mRNA's 'code' into proteins. Molecules called ribosomes carry out this process, building up proteins from the amino acids coded for.

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Nucleotides

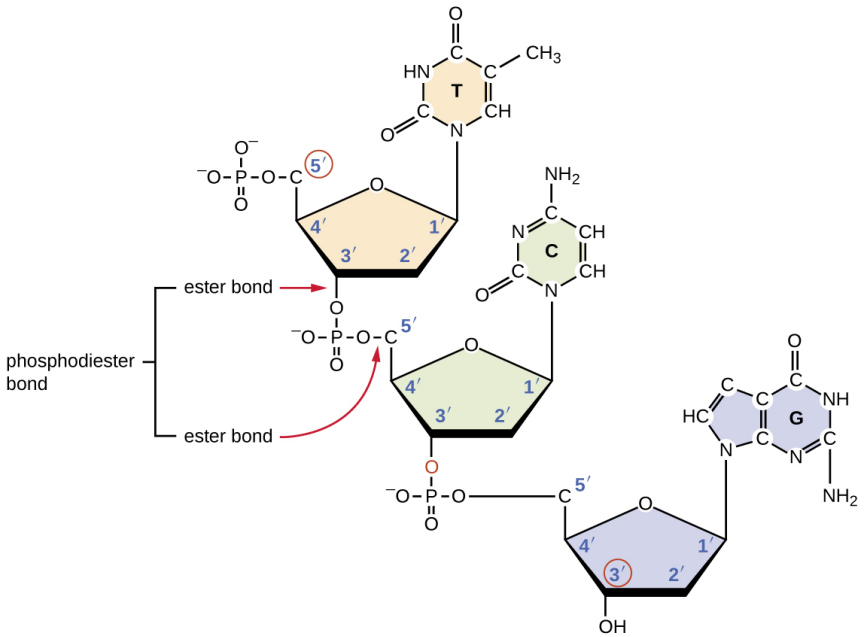
- DNA and RNA is made of nucleotides
- Each nucleotide (Fig. a): one or more phosphate groups, a pentose sugar (c), and a nitrogen-containing base (b)
- DNA bases: adenine (A), cytosine (C), guanine (G), thymine (T)
- RNA: like DNA, but ribose instead of deoxyribose and uracil (U) replaces T!



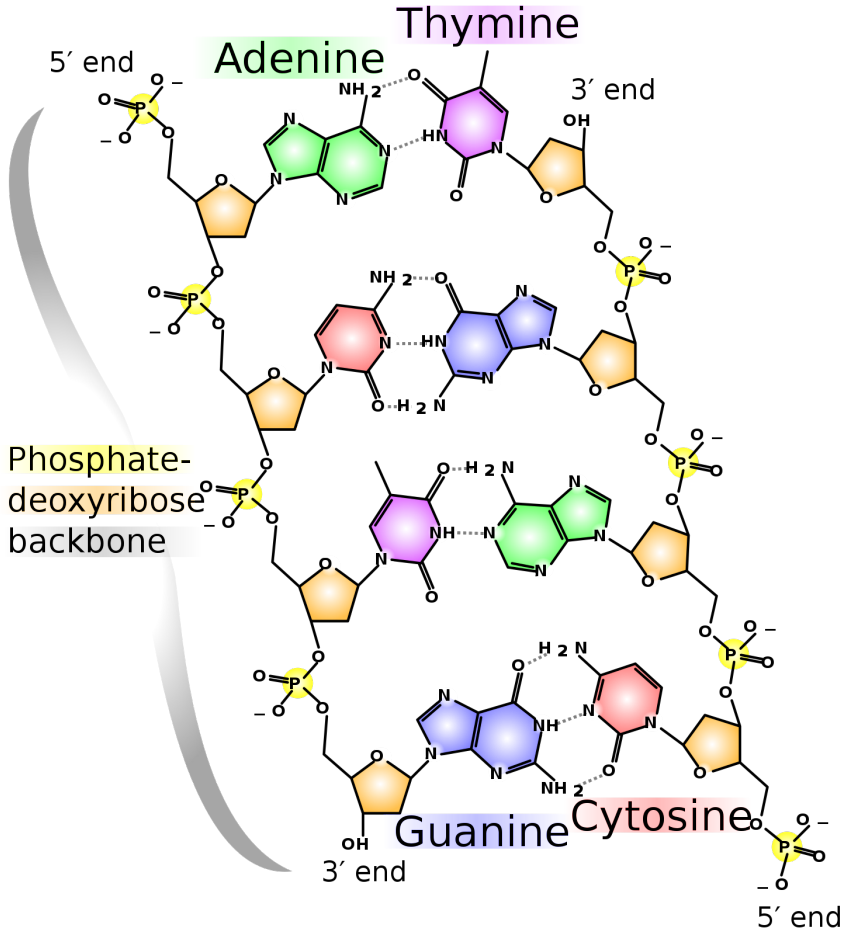
(from: https://philschatz.com/anatomy-book/contents/m46008.html, but corrected)

DNA molecules

- DNA is a polymer (composed of monomers, the nucleotides)
- The phosphate group and the sugar constitute the “backbone” of the molecule (bases are attached to it)
- Thus, the genetic information contained in the DNA is codified by a four-letter alphabet (A,C,G,T)!



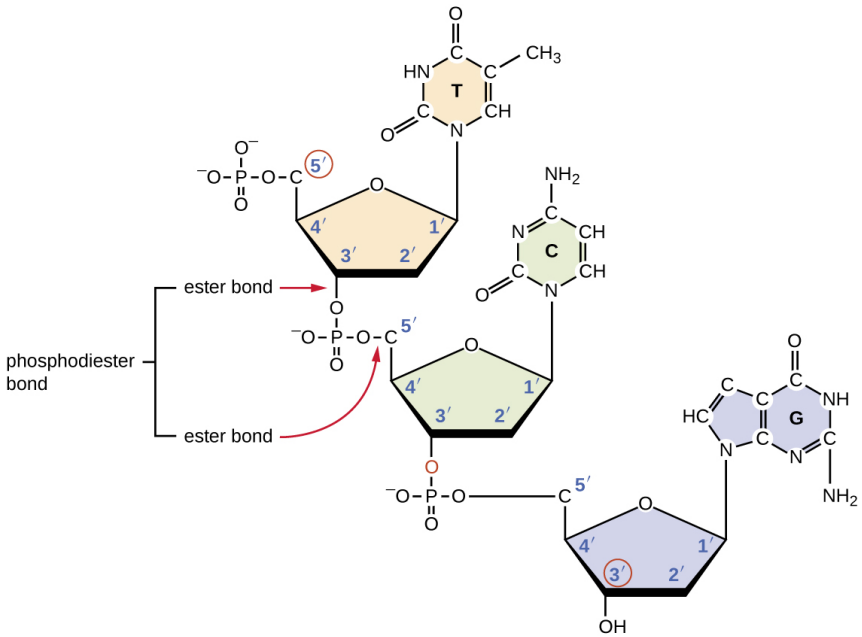
(from: Parker et al., 2021)



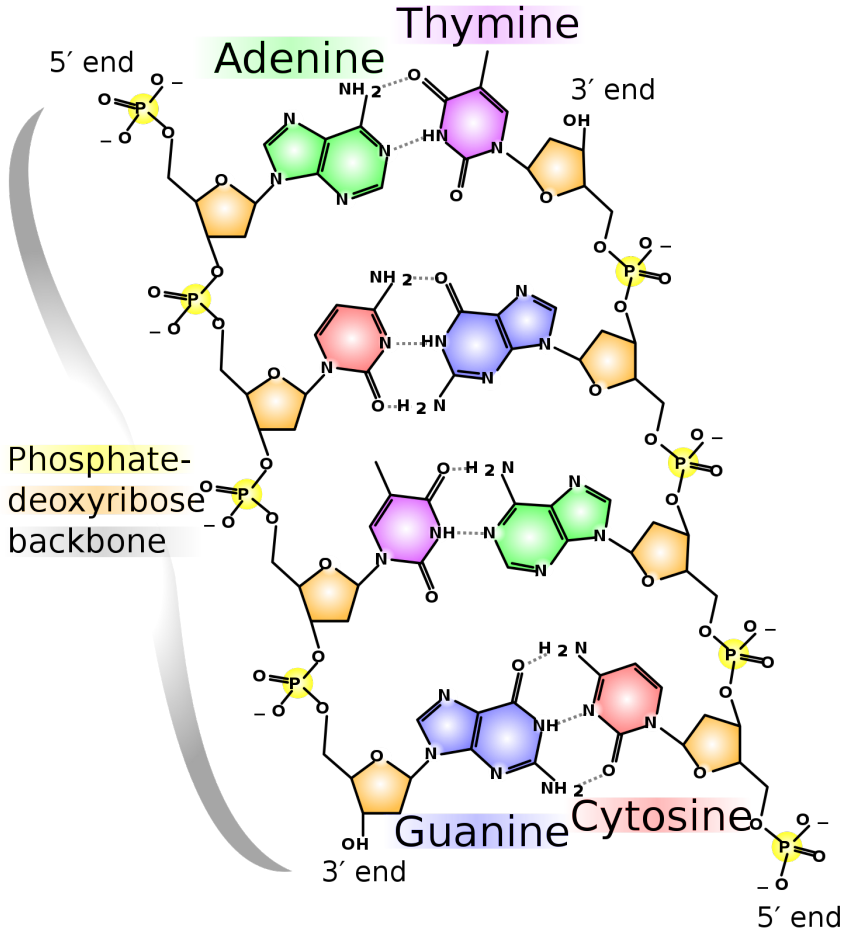
(from: Wikidoc, author: Marshallsumter)

DNA molecules

- Backbone formed by covalent bonds:
  The phosphate group attached to the 5′ carbon of the sugar of one nucleotide is linked to the hydroxyl (OH) group of the 3′ carbon of the sugar of the next nucleotide
- **Single strand**: read from 5′ to 3′
- **Double helix**: the two strands have the opposite orientation (↓↑) !



(from: Parker et al., 2021)



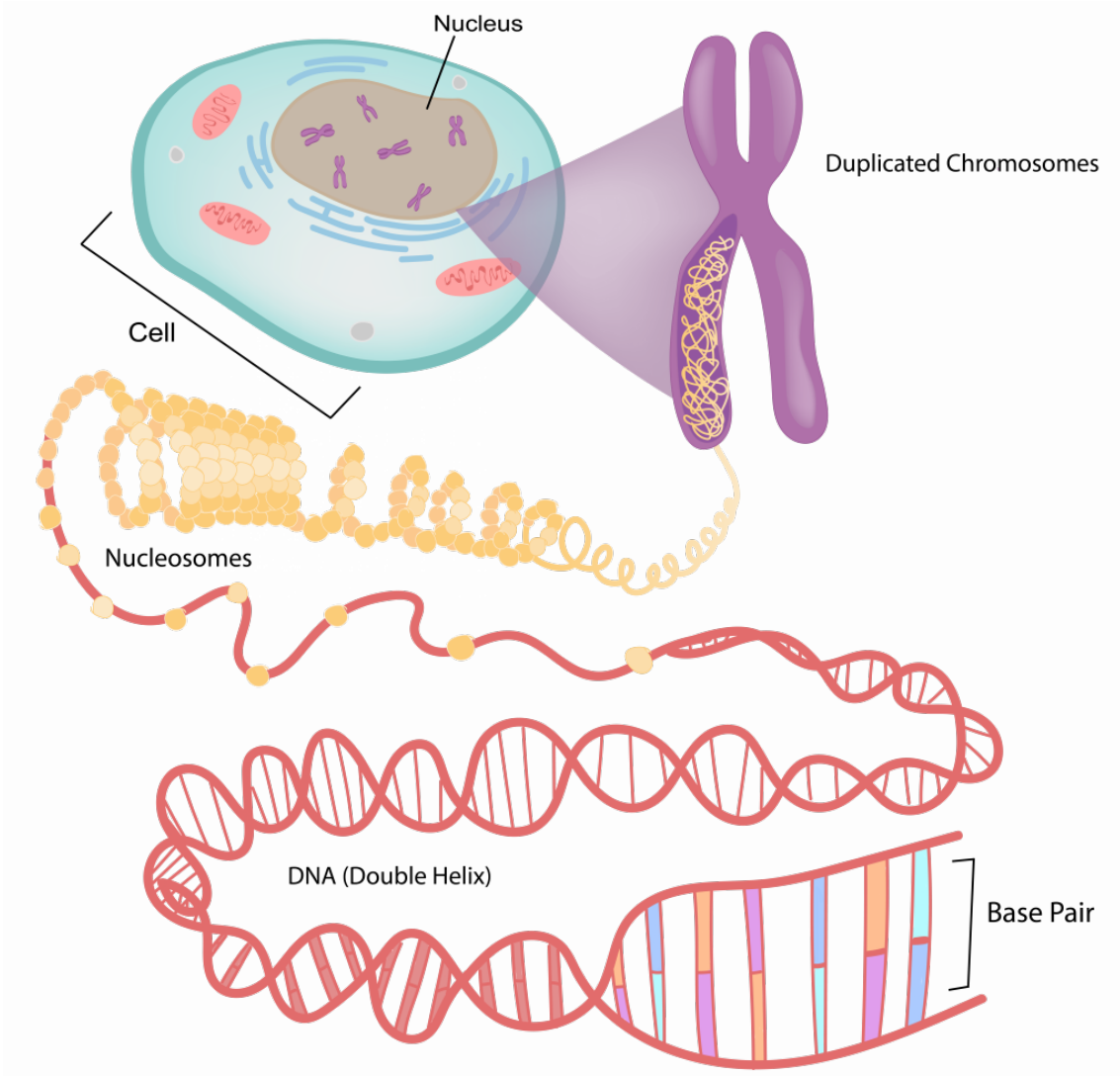
(from: Wikidoc, author: Marshallsumter)

- **Double helix**: A pairs with T and C pairs with G (“base pairs”, bp)!

Chromosomes

DNA is organized into chromosomes

(with exceptions: prokaryotes usually have a single circular genome)



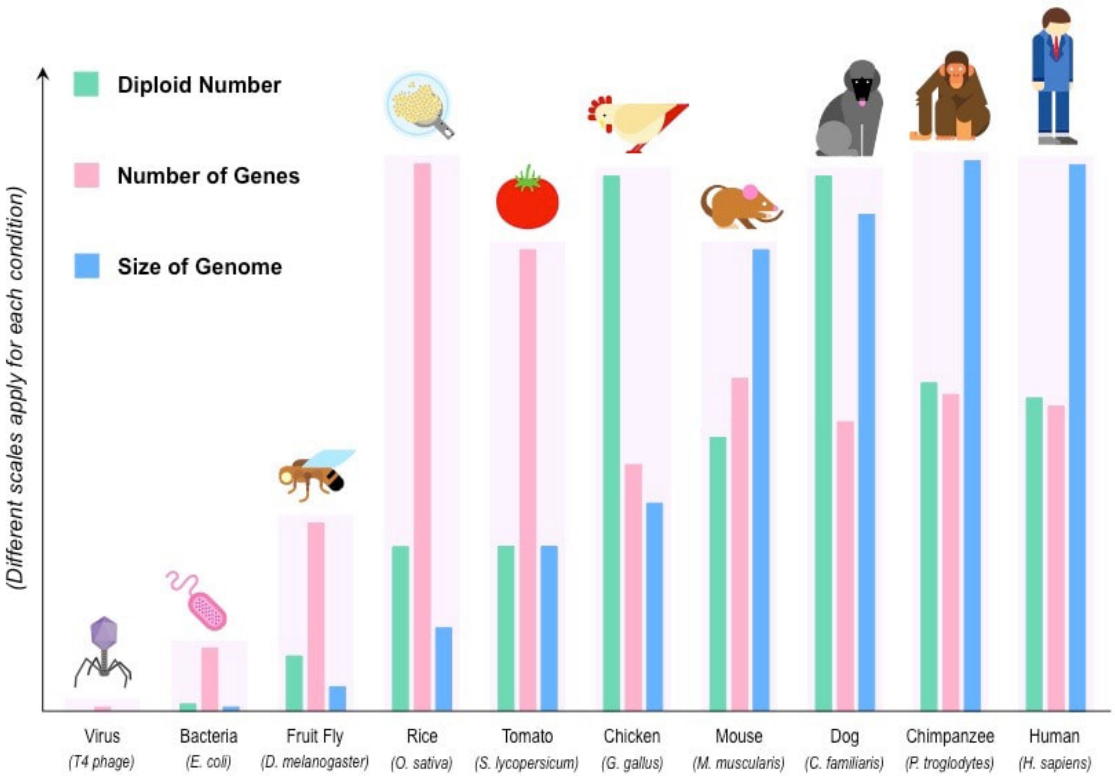
(from: Cotner and Wassenberg, 2020)

Genome size

Diploid numbers (# of chromosomes) and genome sizes differ widely between species

Species	*Parascaris equorum*	*Oryza sativa*	*Homo sapiens*	*Pan troglodytes*	*Canis familiaris*
Chromosome #	4	24	46	48	78
Common Name	Roundworm	Rice	Human	Chimpanzee	Dog

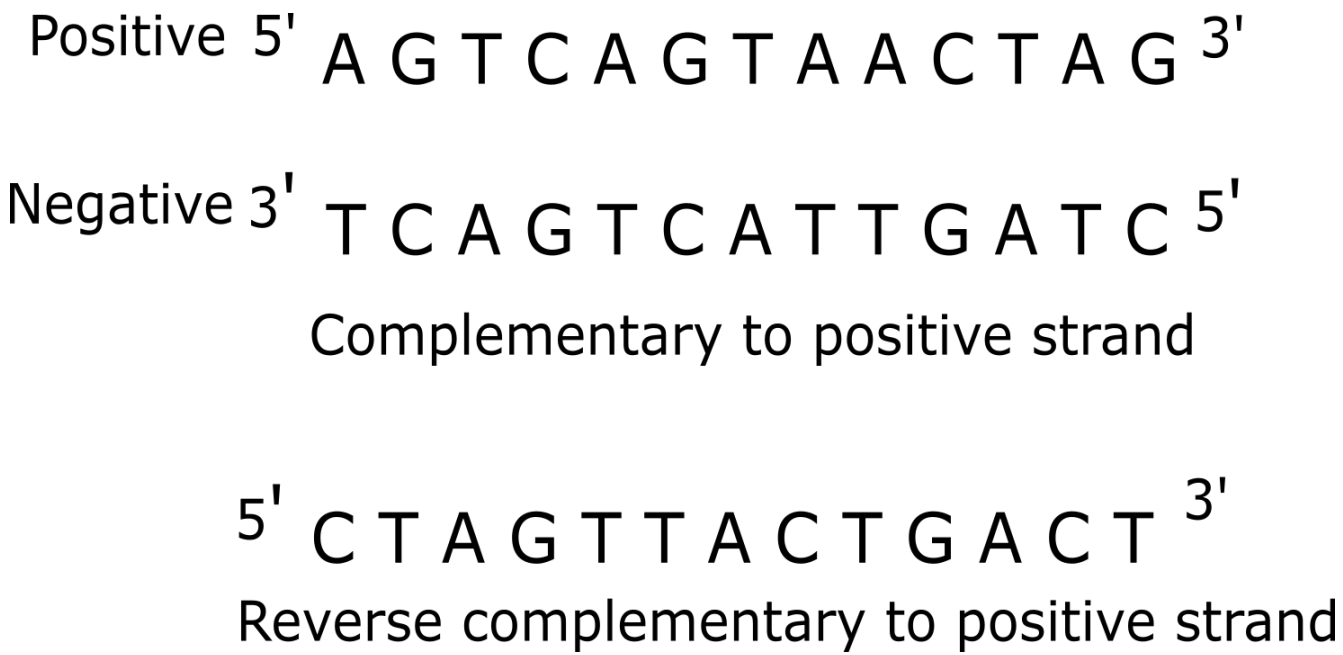
Human: 23 chromosome pairs (1...22, XX or XY) $\rightarrow$ diploid number 46



Human: $\sim$3.1 billion bp

(images from Bioninja)

Reverse complement



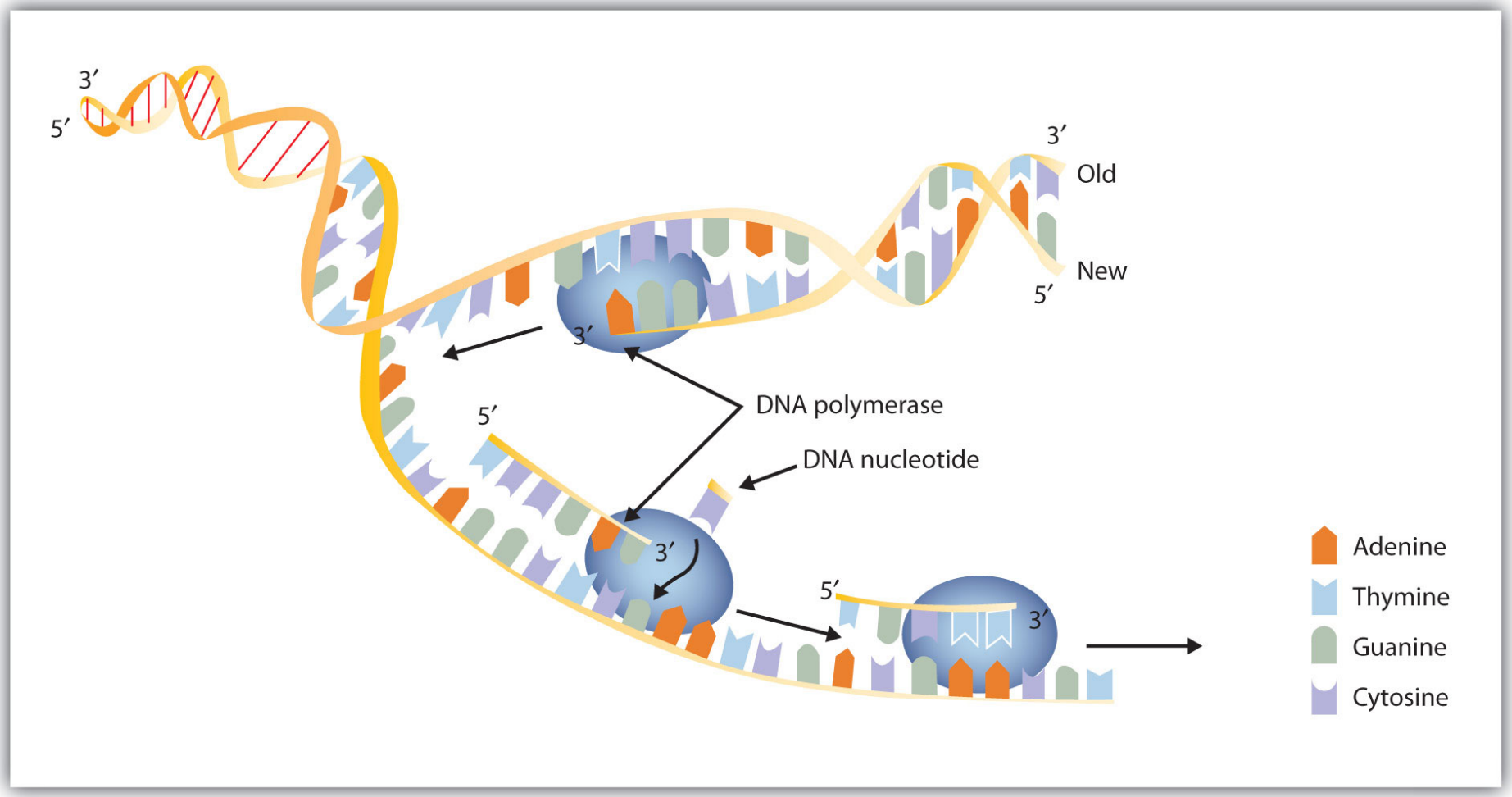
1. Complement: replace A by T, C by G, G by C, and T by A
2. Reverse complement: complement *and* reversed order!

If on one DNA strand you have the sequence "`AGTCAGTAACTAG`", the opposite strand has the sequence "`CTAGTTACTGACT`" (reverse complement!)

DNA replication

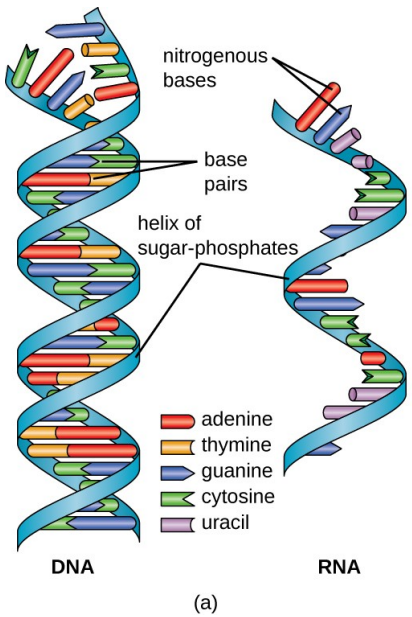
DNA replication (for cell division, etc.) is done by reconstructing the opposite strand for each of the two separated original DNA strands!

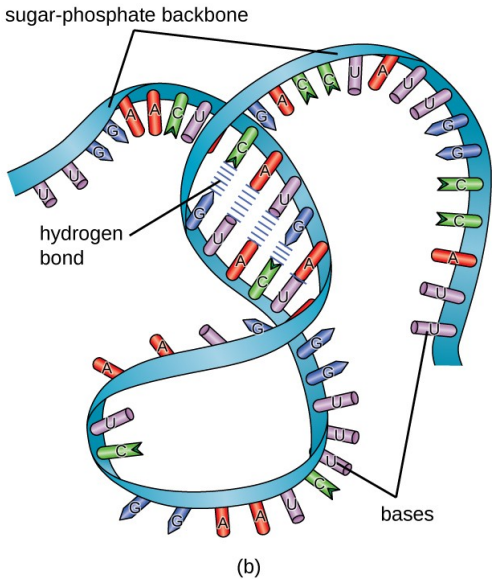
- Biochemical extension of DNA molecules by the DNA polymerase always from $5'$ to $3'$!



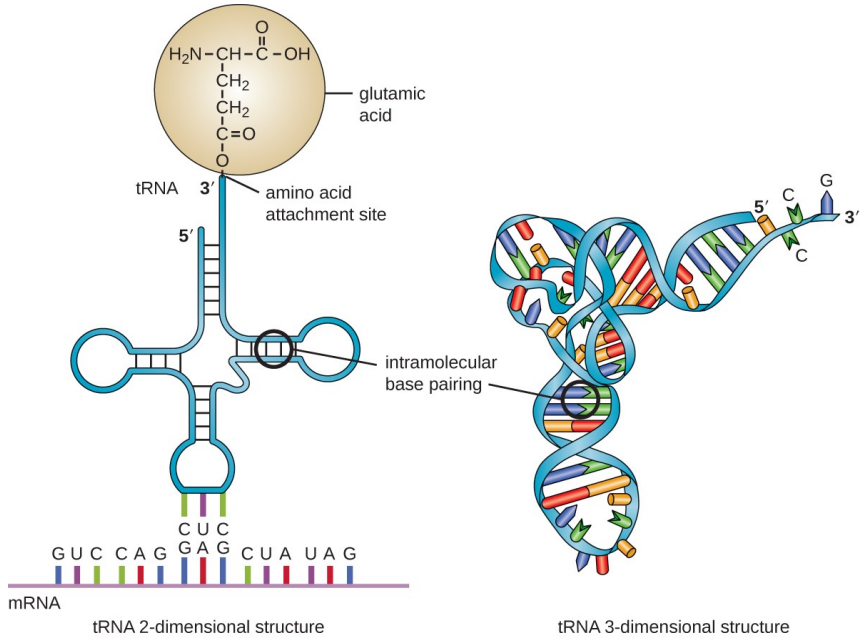
(from: Bennett, 2016)

RNA molecules





(from: Parker et al., 2021)



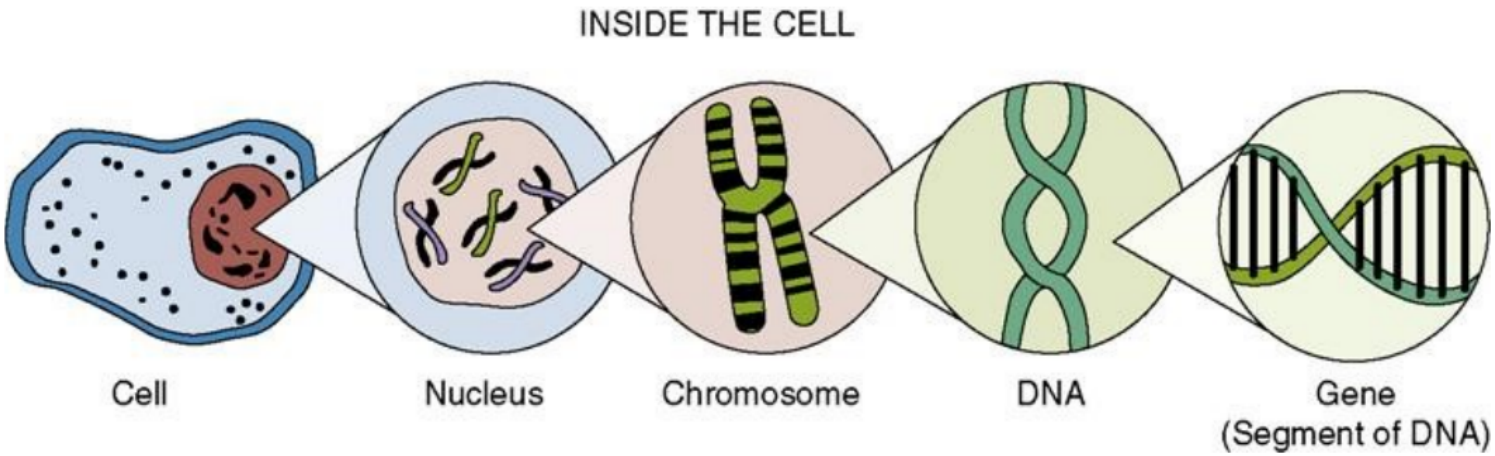
(tRNA; from: `https://courses.lumenlearning.com`)

- RNA nucleotides: like DNA, but ribose instead of deoxyribose and uracil (U) replaces T!
- RNA molecules are *single-stranded*!
- Base pairing is still possible $\rightarrow$ secondary structure!

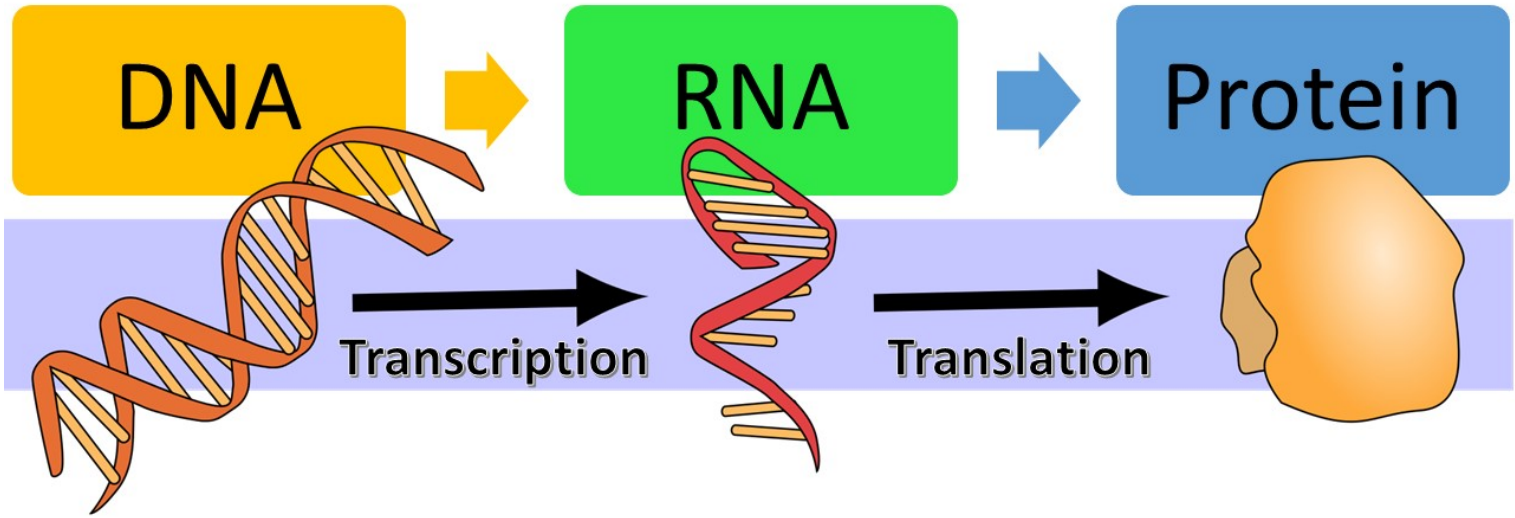
- Transfer RNA (tRNA)
- Ribosomal RNA (rRNA)
- Messenger RNA (mRNA)
- MircoRNA (miRNA)
- Long non-coding RNA (lncRNA)
- Small nuclear RNA (snRNA)
- Small nucleolar RNA (snoRNA)

Encoding and decoding life

- Every living organism is encoded by one or more DNA molecules, that are continuously read and interpreted by its cells
- Questions:
  - Where is the DNA located in the cell?
    - ★ Prokaryotes (bacteria, cyanobacteria): “free-floating” in the cytoplasm
    - ★ Eukaryotes (animals, plants, fungi): in the cell nucleus, “packaged” into chromosomes
  - How much and what kind of information contains a living cell?
  - How is that information encoded, stored, transmitted and decoded by DNA?



The “Central Dogma” of molecular biology



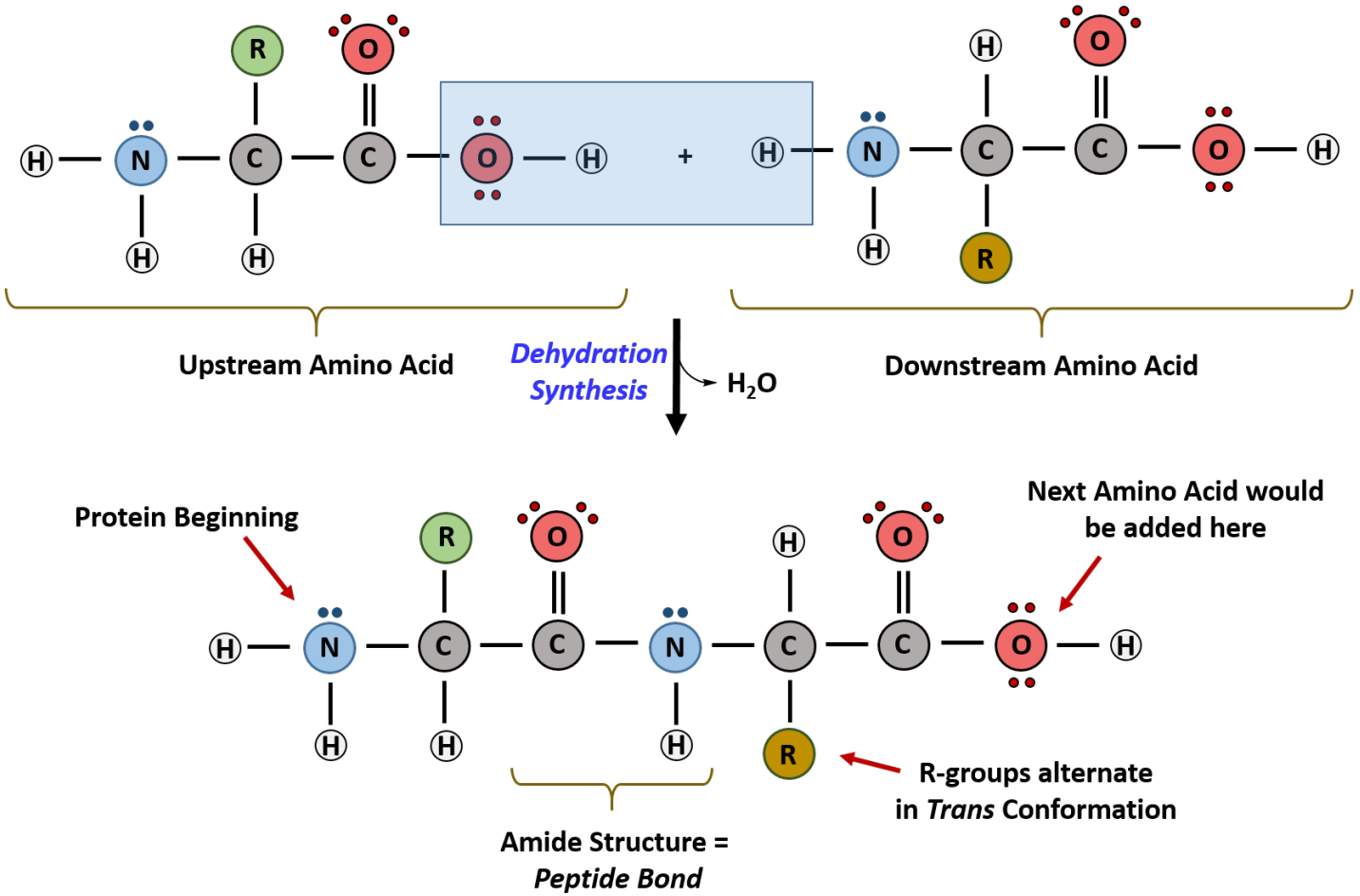
(from: Bierema, 2021)

- DNA stores the information that determines a protein’s structure
  - DNA is “transcribed” (i.e., copied) into RNA
- RNA mediates the transformation of genetic information into functional molecules
  - “Messenger RNA” (mRNA) is “translated” into a protein
- Proteins exert a biological function (depends on structure & chemical properties)

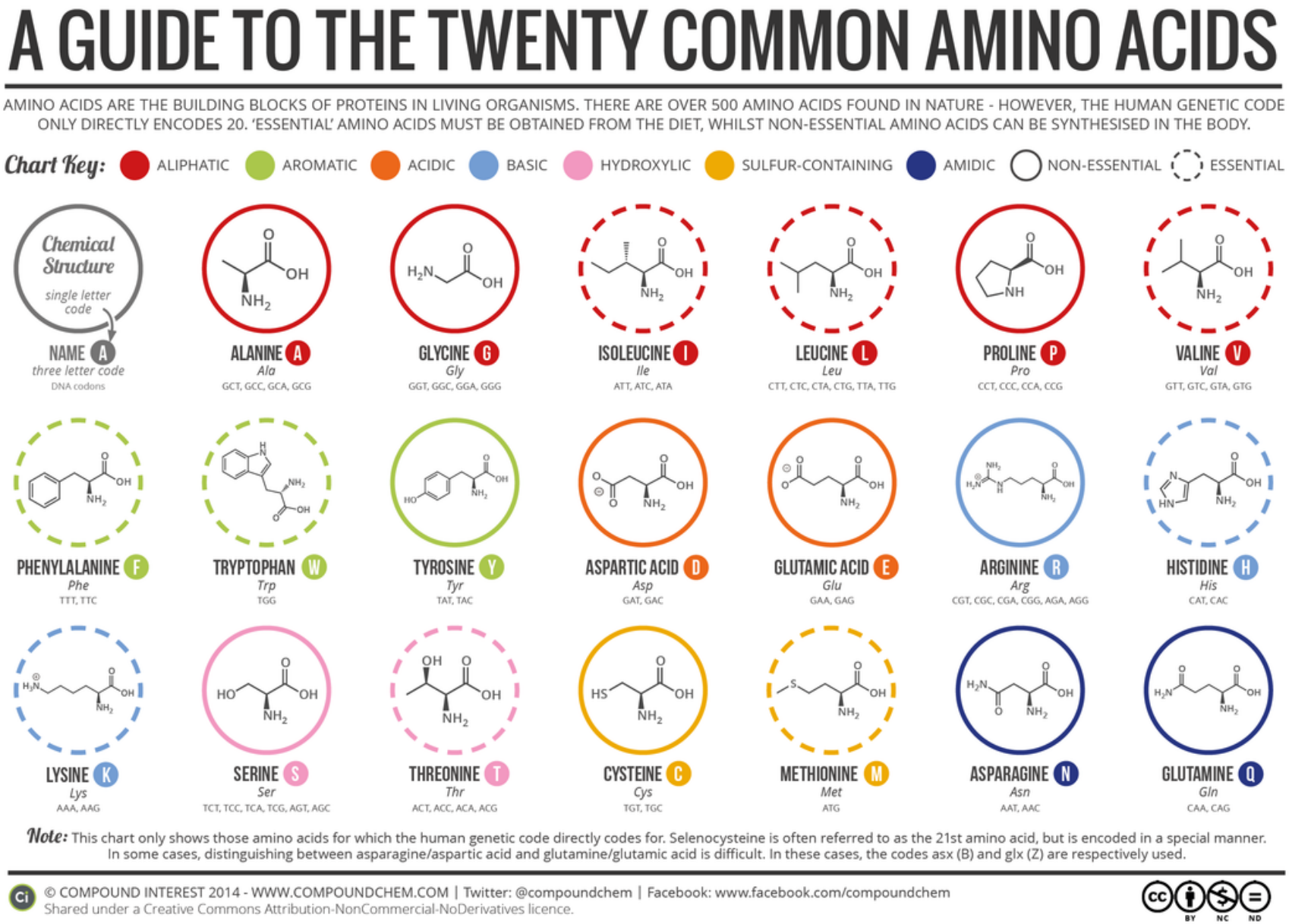
- Note: this holds only for protein-coding genes
  - Non-coding RNAs are functional molecules which need not be translated
- Note: this is a very simplified view
  - E.g., one protein-coding gene can “code for” multiple proteins

Proteins

- Proteins are polypeptides (chains of indidvidual amino acids)
- Basic amino acid structure:
  amino group ($-NH_3^+$), R group (carbon with side chain R), carboxyl group ($-CO_2^-$)
- Orientation of an amino acid chain from “N-terminus” to “C-terminus”



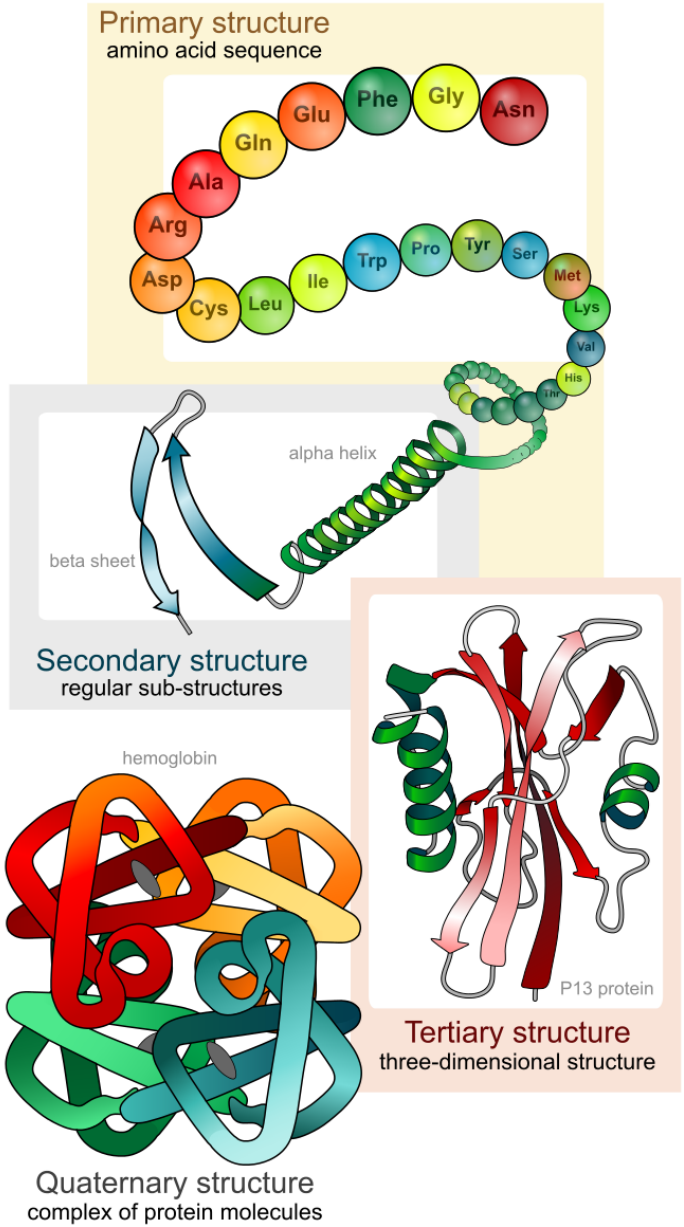
(from: Jakubowski, 2021)



Three-letter code: e.g., "Gly" for glycine; example sequence: Gly-Ser-Tyr-...

One-letter code: e.g., "G" for glycine; example sequence: GSY...

Protein structure

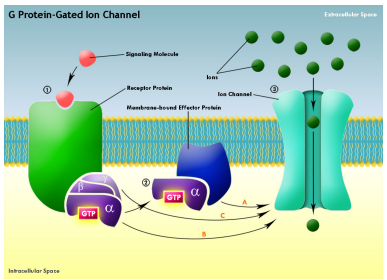


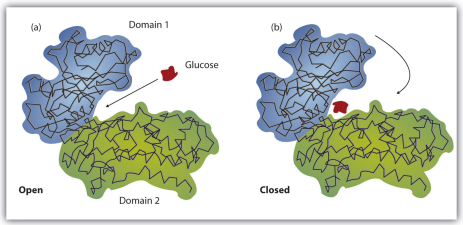
(from: Bartee et al., 2021)

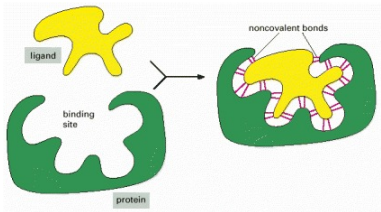
Protein functions

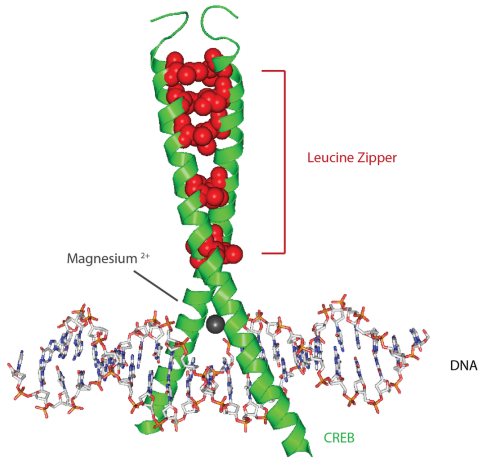
Proteins are involved in most of the tasks essential for life:

- Structural proteins
  - basic building blocks of cells
- Receptor and channel proteins
  - transmembrane proteins which can relay signals into the cell or shuttle substances between compartments of the cell or the cell and the extracellular environment
- Signaling proteins
  - relay signals within the cell
- Enzymes
  - catalyze chemical reactions necessary for the energy metabolism
- Transcription factors
  - bind to DNA to control gene transcription (e.g., gene expression levels)



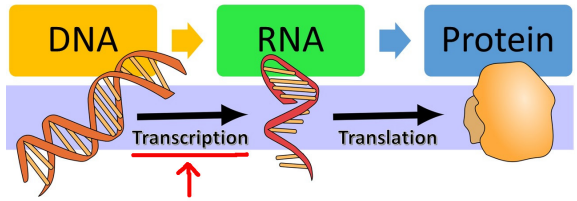




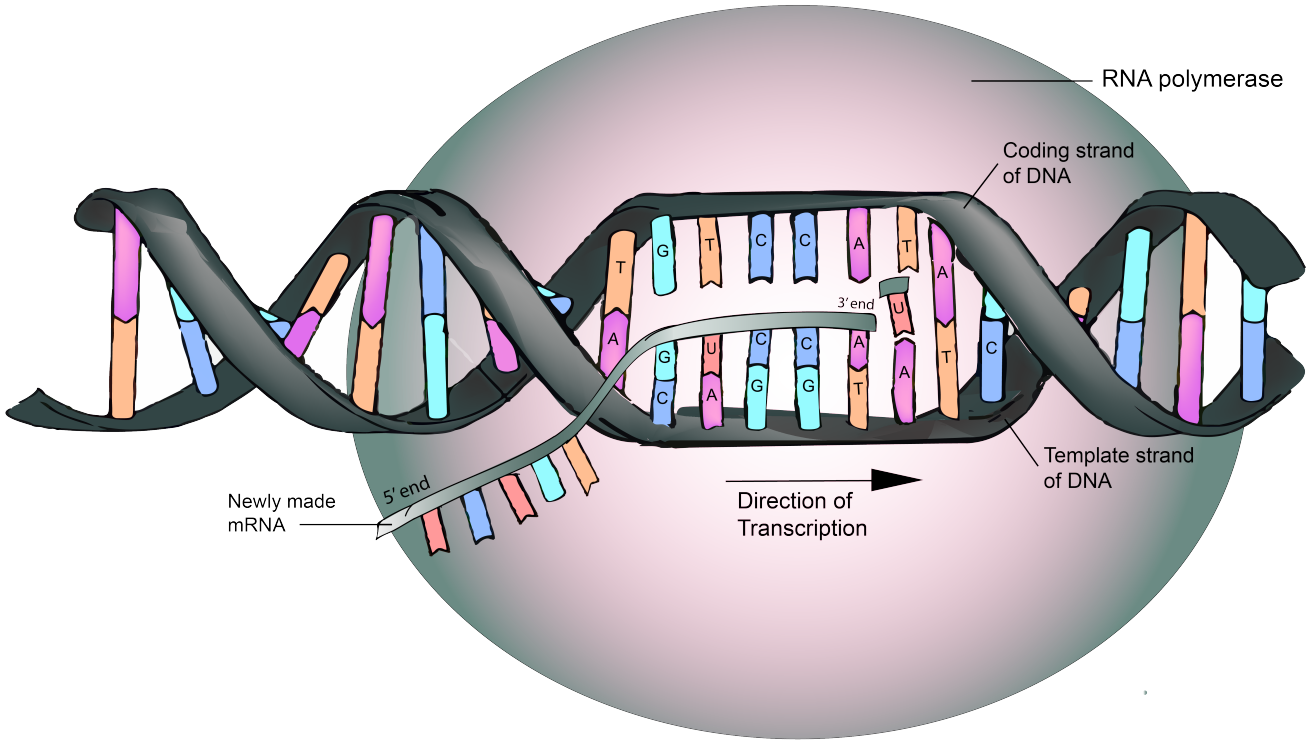


(from: Wikipedia; Bennett, 2016; Alberts et al., 2002; Wikipedia, user: Studentne)

Transcription

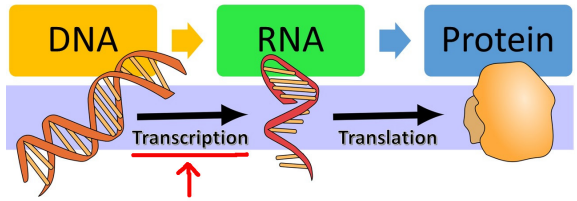


- One strand of DNA is copied into a (reverse) complementary RNA molecule
  - In prokaryotes (without nucleus): messenger RNA (mRNA)
  - In eukaryotes (with nucleus): pre-mRNA (produced *in the nucleus*)

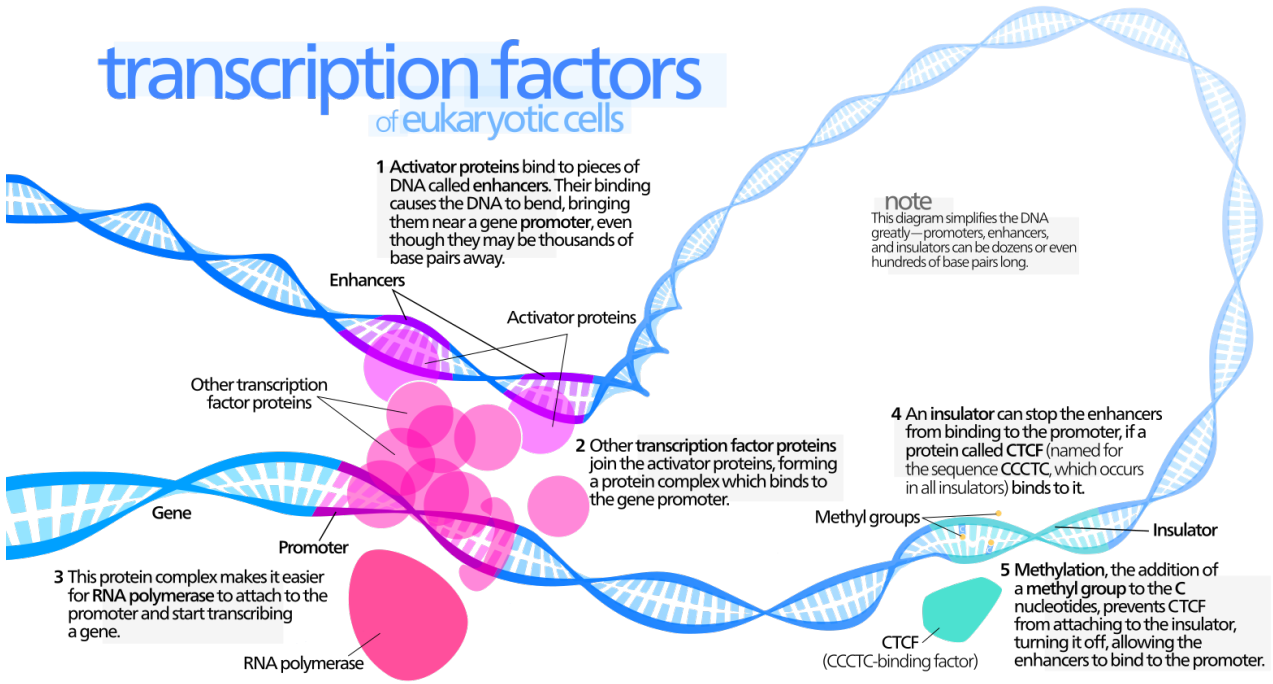


(from: Cotner and Wassenberg, 2020)

Transcription



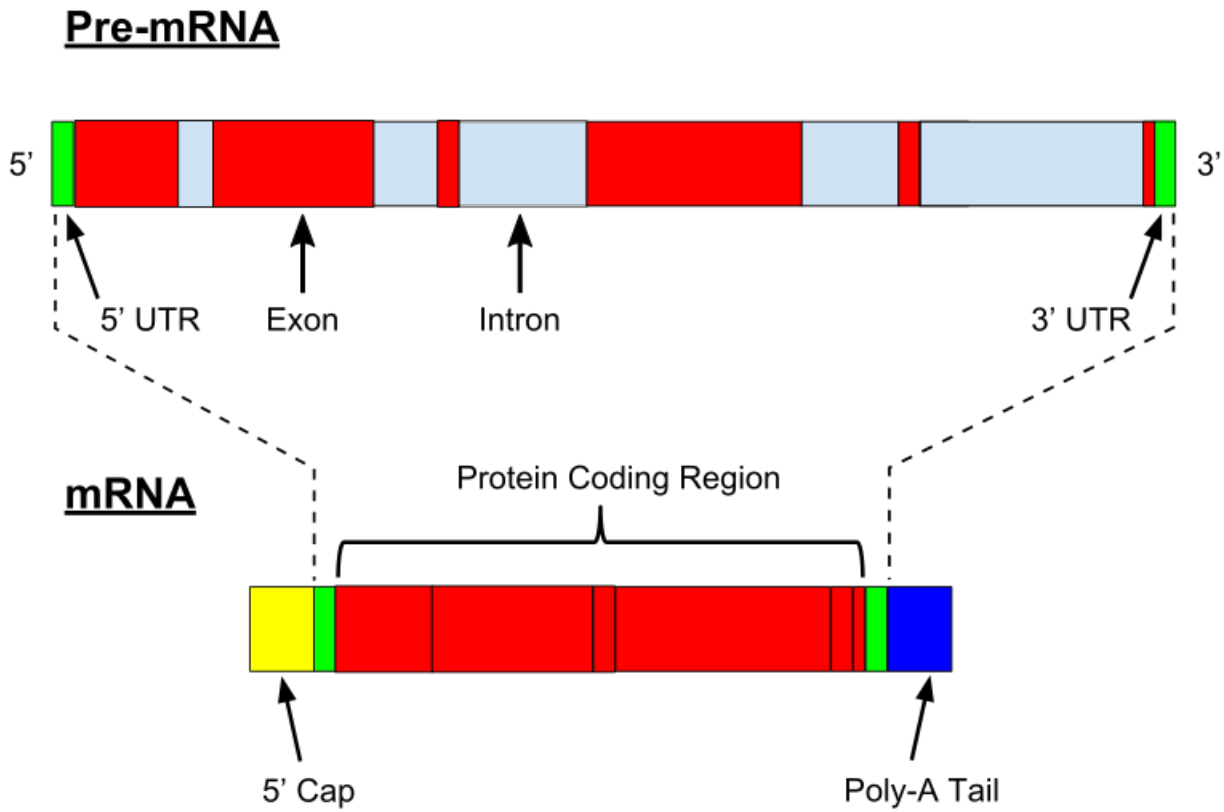
- DNA is copied into RNA by an RNA polymerase (usually RNA polymerase II)
  - Transcription start is aided by transcription factors (TFs) bound on the “promoter” of the gene (around the “transcription start site”; TSS)
  - Transcription start can be aided by TFs bound to a distant “enhancer”



(from: Boundless, 2020)

RNA splicing

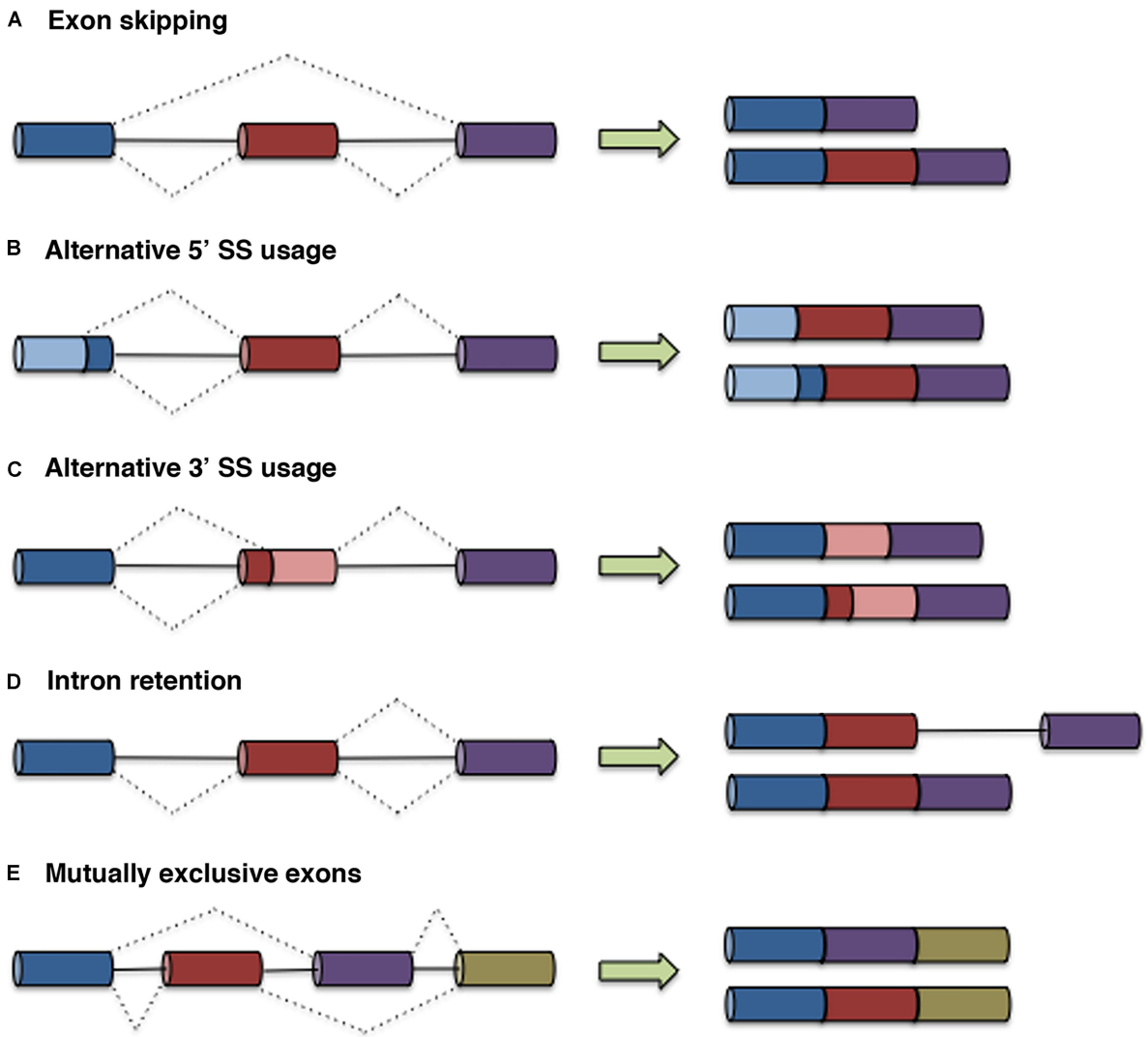
- In eukaryotic cells, transcribed RNA is often spliced, before it is exported from the nucleus to the cytoplasm!
  - "Introns" (grey) are "spliced out" (biochemically removed)
  - "Exons" (green and red) remain to form the mature mRNA
    - ★ A protein-coding RNA begins ($5'$) and ends ($3'$) with "untranslated regions" (UTRs)
    - ★ Attention; often claimed, but **wrong**: exons $\equiv$ coding (translated) regions
  - A $5'$-cap and a $3'$ poly-A tail are often added to the transcript



(from: Wikipedia; users: Nastypatty, Smartse, Manudouz)

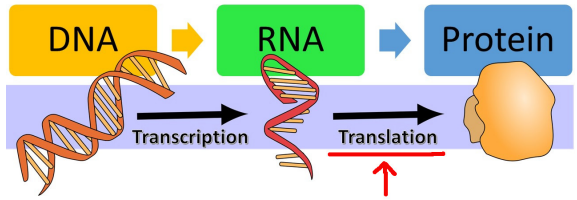
Alternative splicing

- *Alternative splicing* of transcripts allows to produce different gene products (e.g., proteins) from the same gene!

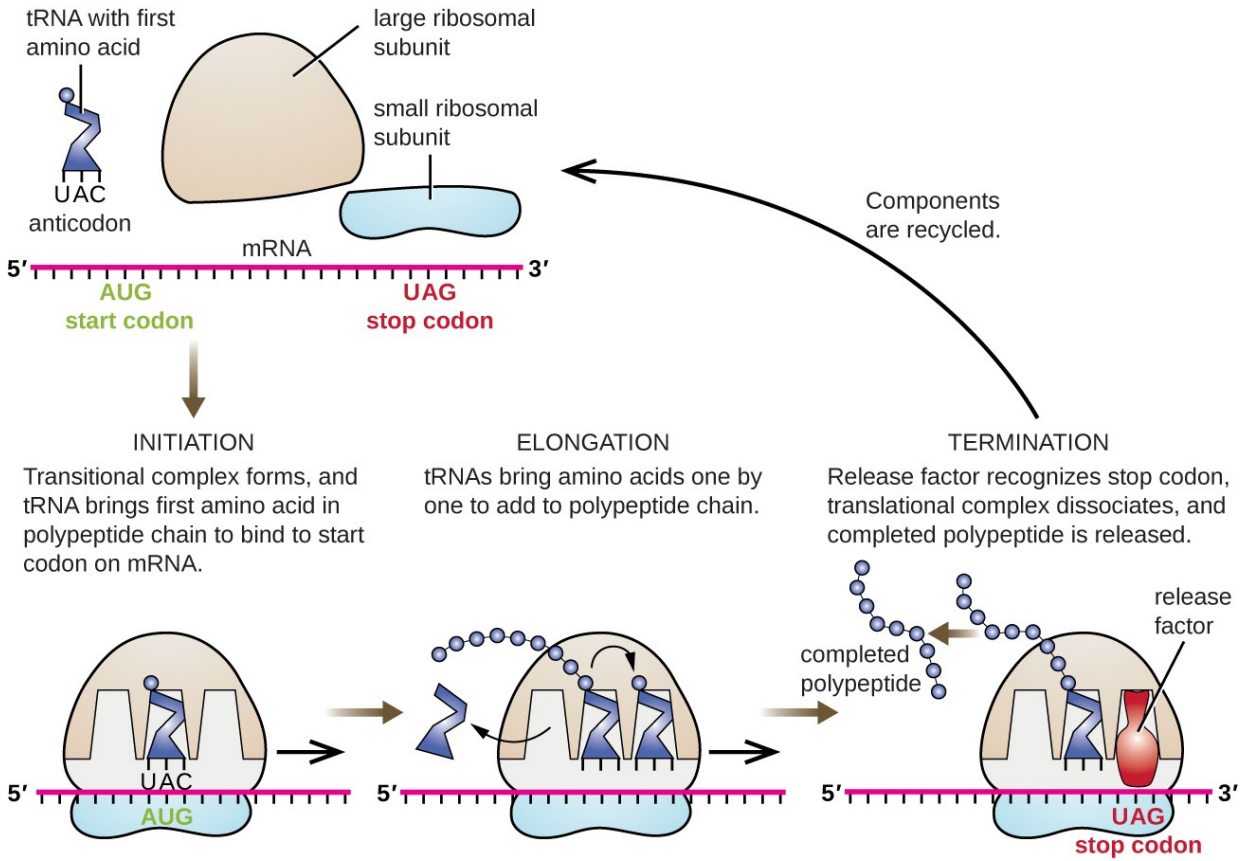


(from: Suñé-Pou et al., 2020; but corrected)

Translation (protein synthesis)

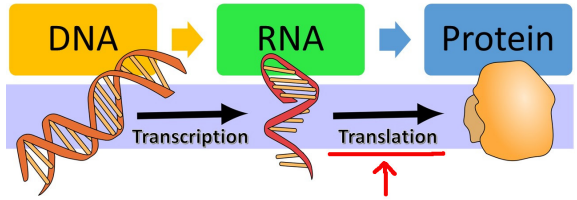


- mRNA transcripts are “translated” into proteins by the ribsosome



(from: `https://courses.lumenlearning.com/microbiology/`)

Translation (protein synthesis)



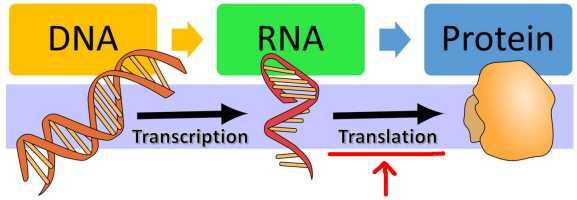
- The genetic code: “codons” (nucleotide triplets) encode amino acids
- Translation starts at an `AUG` and ends at the first `UAA`, `UAG` or `UGA`!

second letter

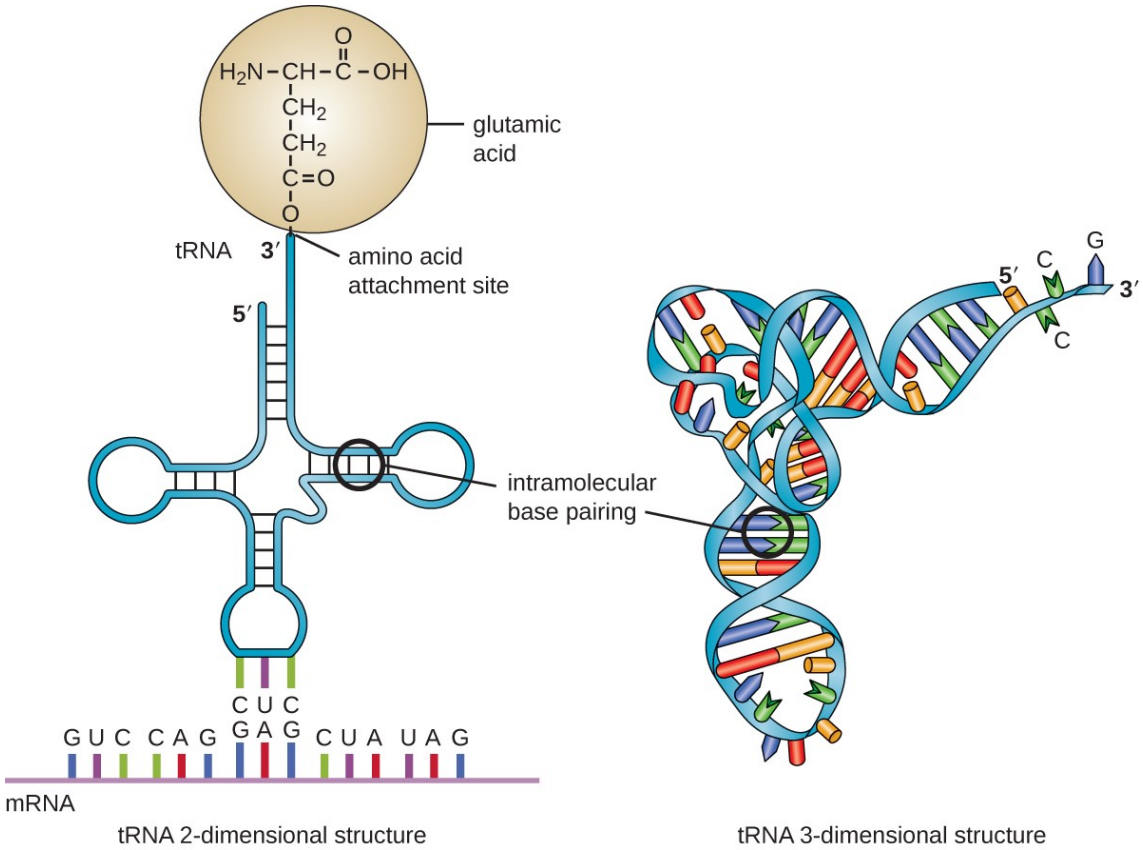
first letter	U	C	A	G	third letter
U	UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr **UAA stop** **UAG stop**	UGU Cys UGC Cys **UGA stop** UGG Trp	**U** **C** **A** **G**
C	CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gln CAG Gln	CGU Arg CGC Arg CGA Arg CGG Arg	**U** **C** **A** **G**
A	AUU Ile AUC Ile AUA Ile **AUG Met**	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg	**U** **C** **A** **G**
G	GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly	**U** **C** **A** **G**

(from: `https://courses.lumenlearning.com/microbiology/`)

Translation (protein synthesis)



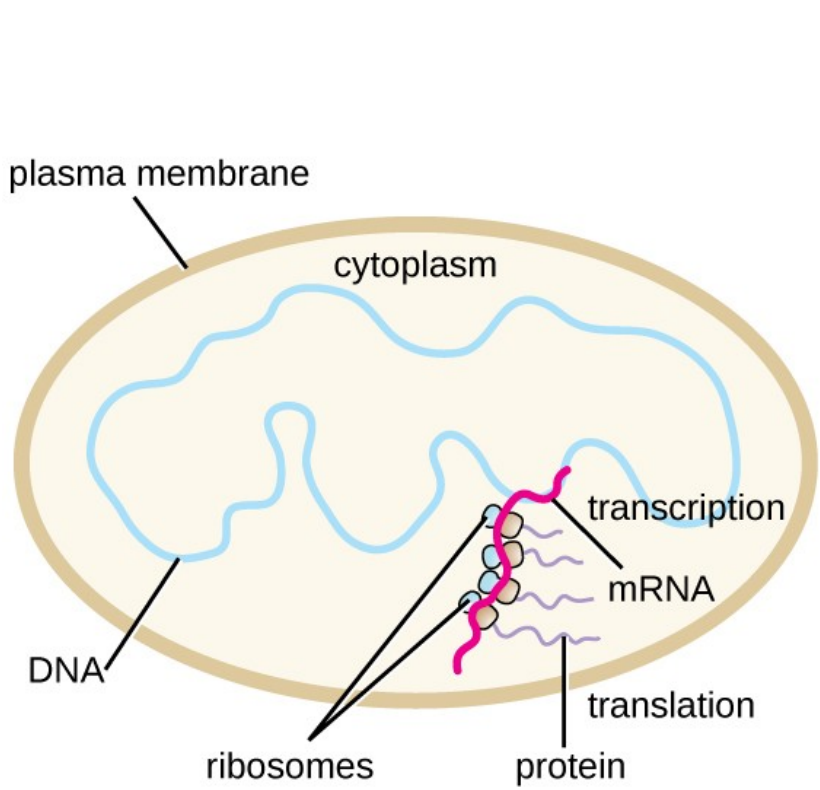
- Each codon is recognized by a tRNA which carries the correct amino acid

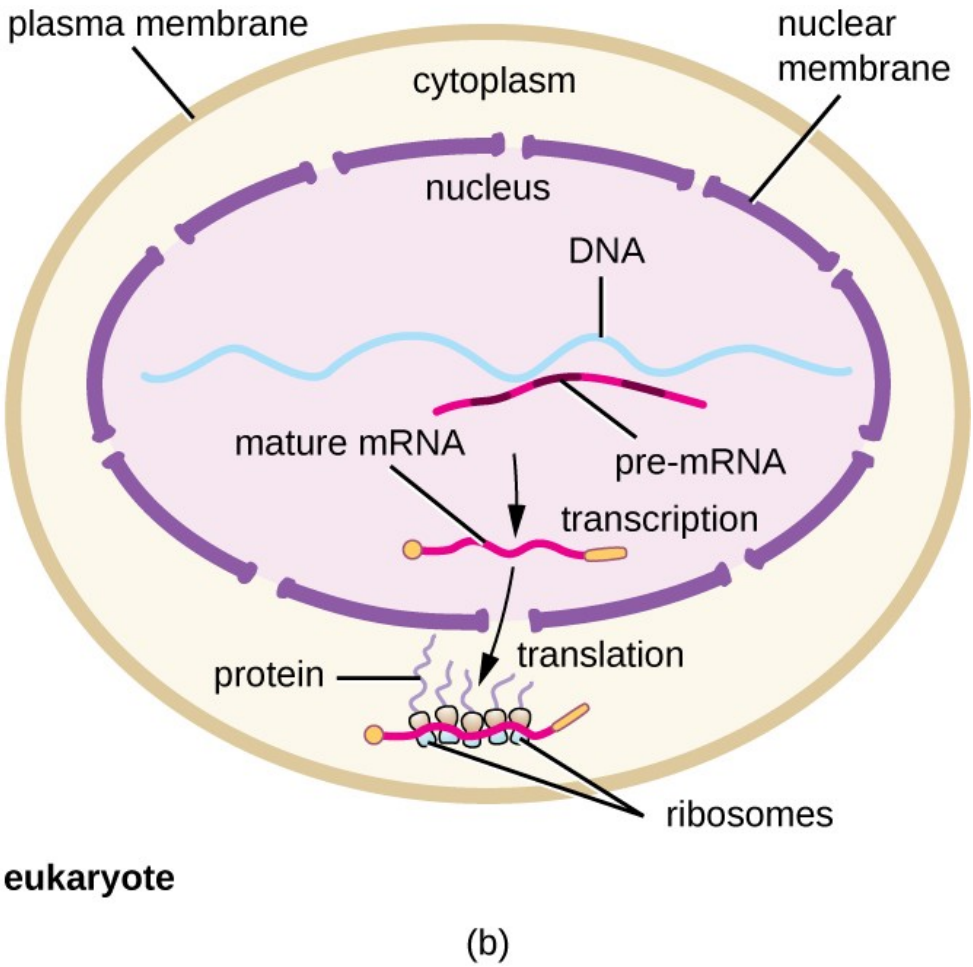


(tRNA; from: `https://courses.lumenlearning.com`)

Prokaryotes vs. eukaryotes

- In eukaryotes: transcription and translation in different compartments
  $\rightarrow$ allows for alternative splicing!

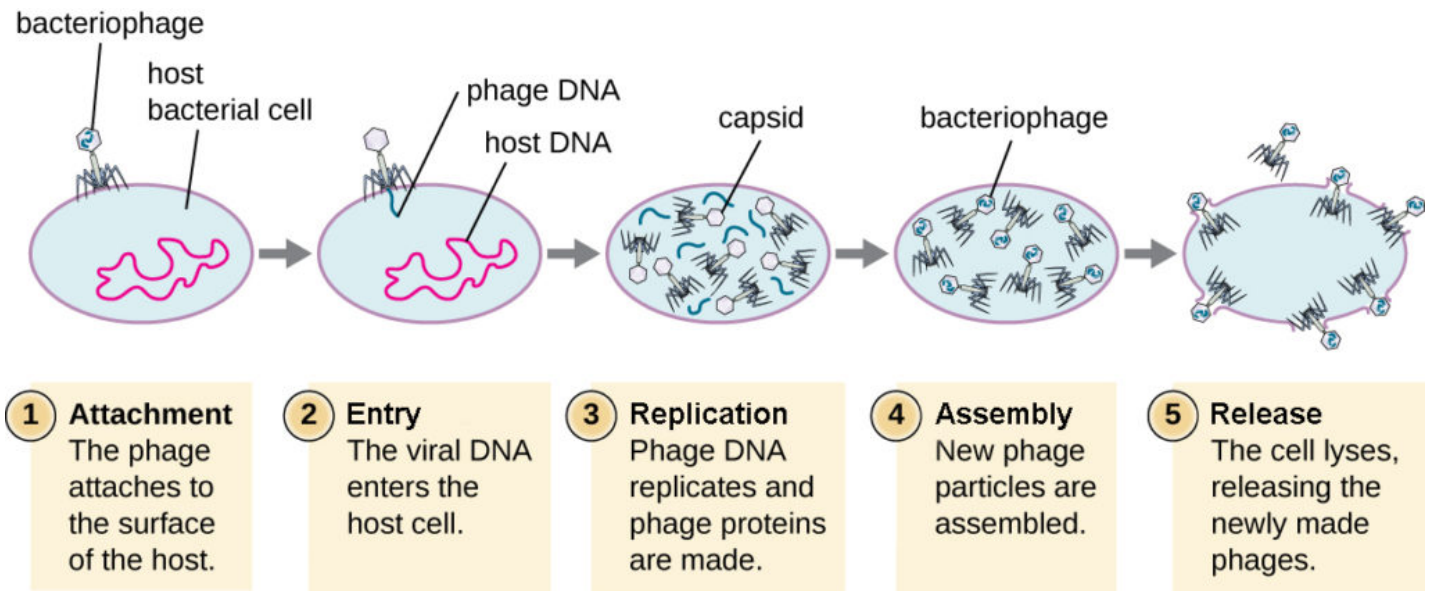


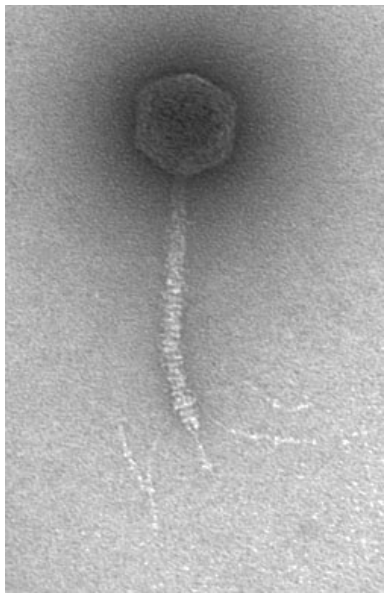


(tRNA; from: `https://courses.lumenlearning.com/microbiology/`)

And viruses?

- Viruses: “organisms at the edge of life”; exploit the host cell for replication!
- The genetic material may be DNA or RNA, single-stranded or double-stranded, linear or circular; some viruses integrate genetic material into the host genome
- The virus envelope/capsule may contain additional viral proteins (e.g., reverse transcriptase, integrase)
- Example: bacteriophage lambda (infects bacteria)

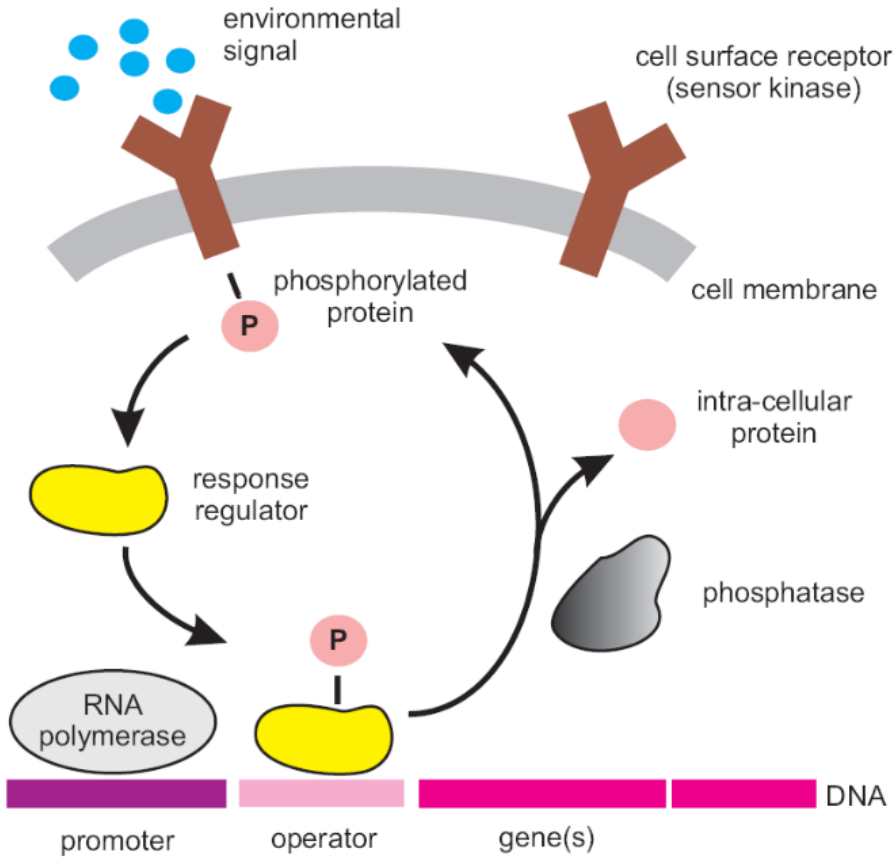




(from: https://courses.lumenlearning.com/ivytech-bio1-1/ and http://evilutionarybiologist.blogspot.com/)

Cellular signaling

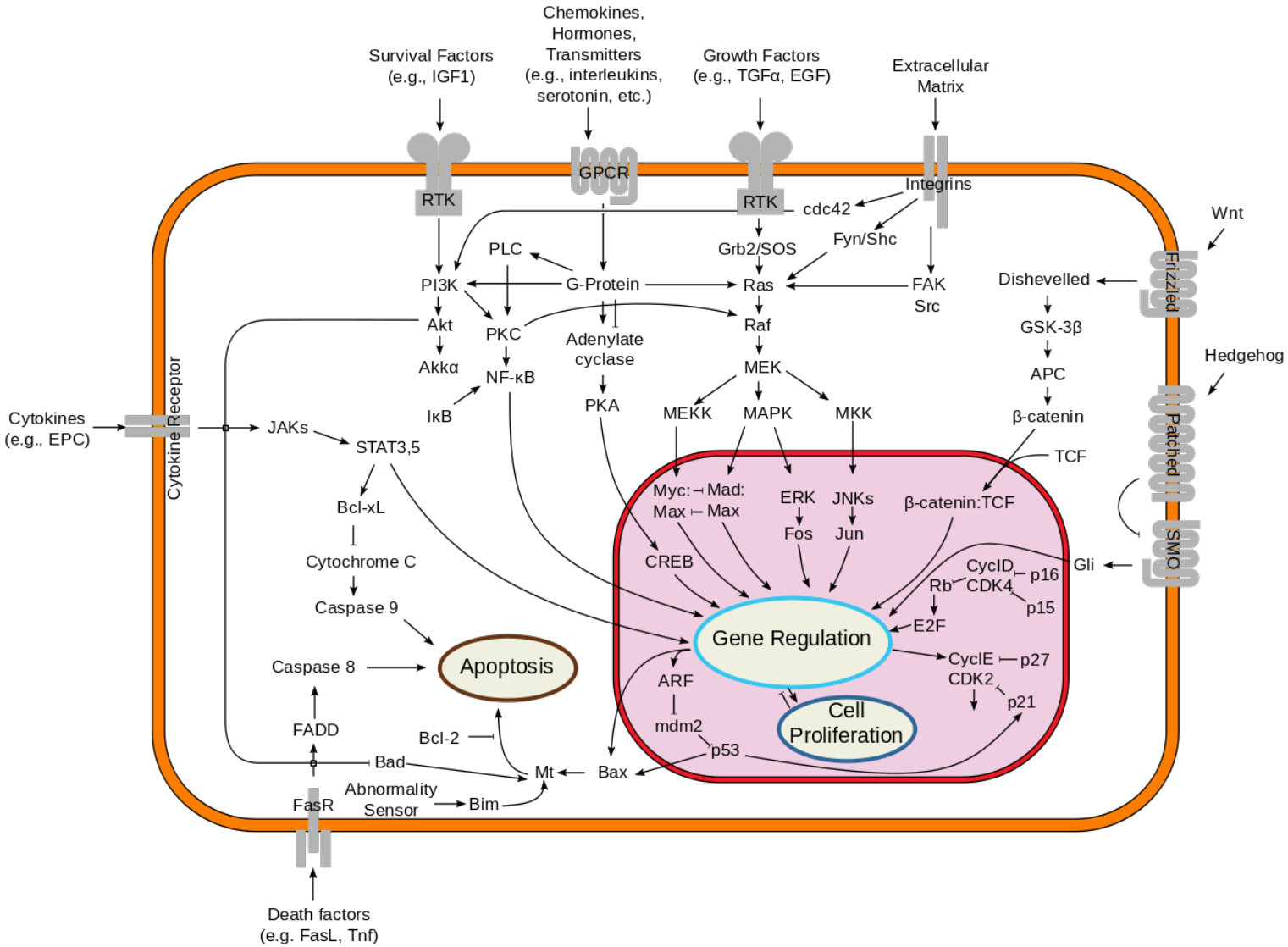
A hypothetical, simple signaling pathway:



http://www.informatik.uni-rostock.de/~lin/GC/Slides/Wolkenhauer.pdf

Cellular signaling

A little more realistic (but still simplified!):

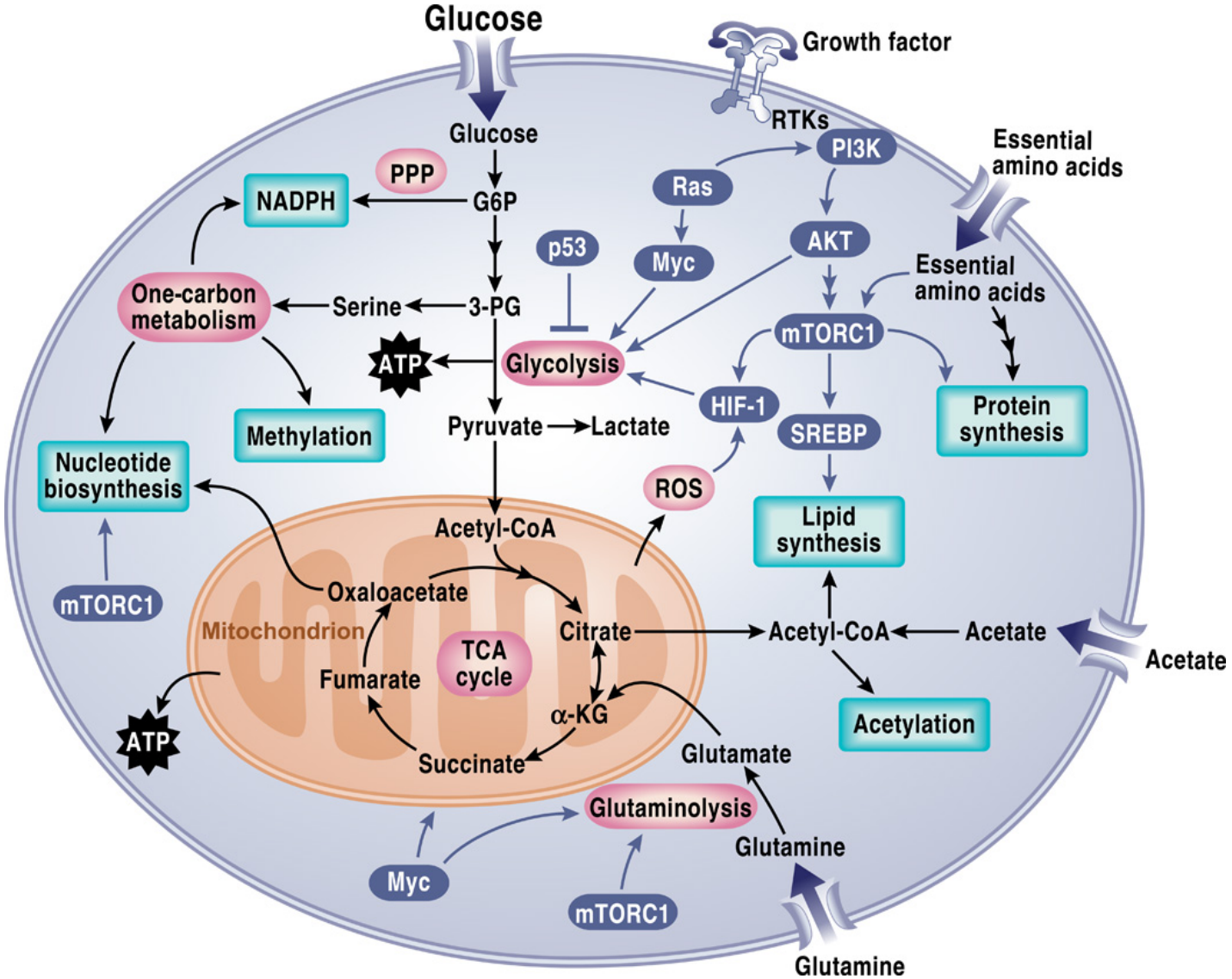


Major signal transduction pathways in mammals

(from: Wikipedia, user: Cybertory)

Cellular signaling

A little more realistic (but still simplified!):

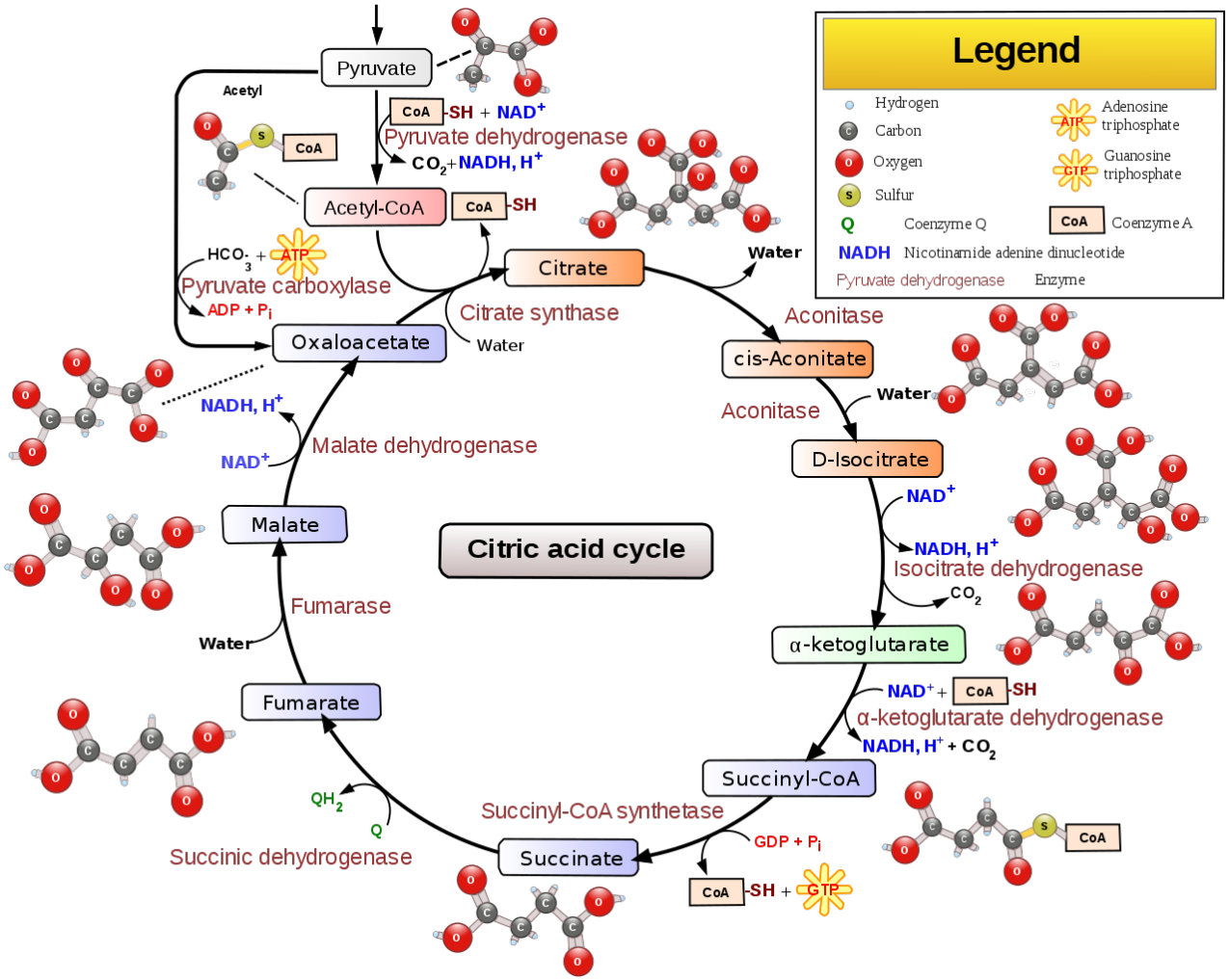


Signaling pathways that regulate cancer metabolism

(from: Deberardinis and Chandel, 2016)

Metabolic processes

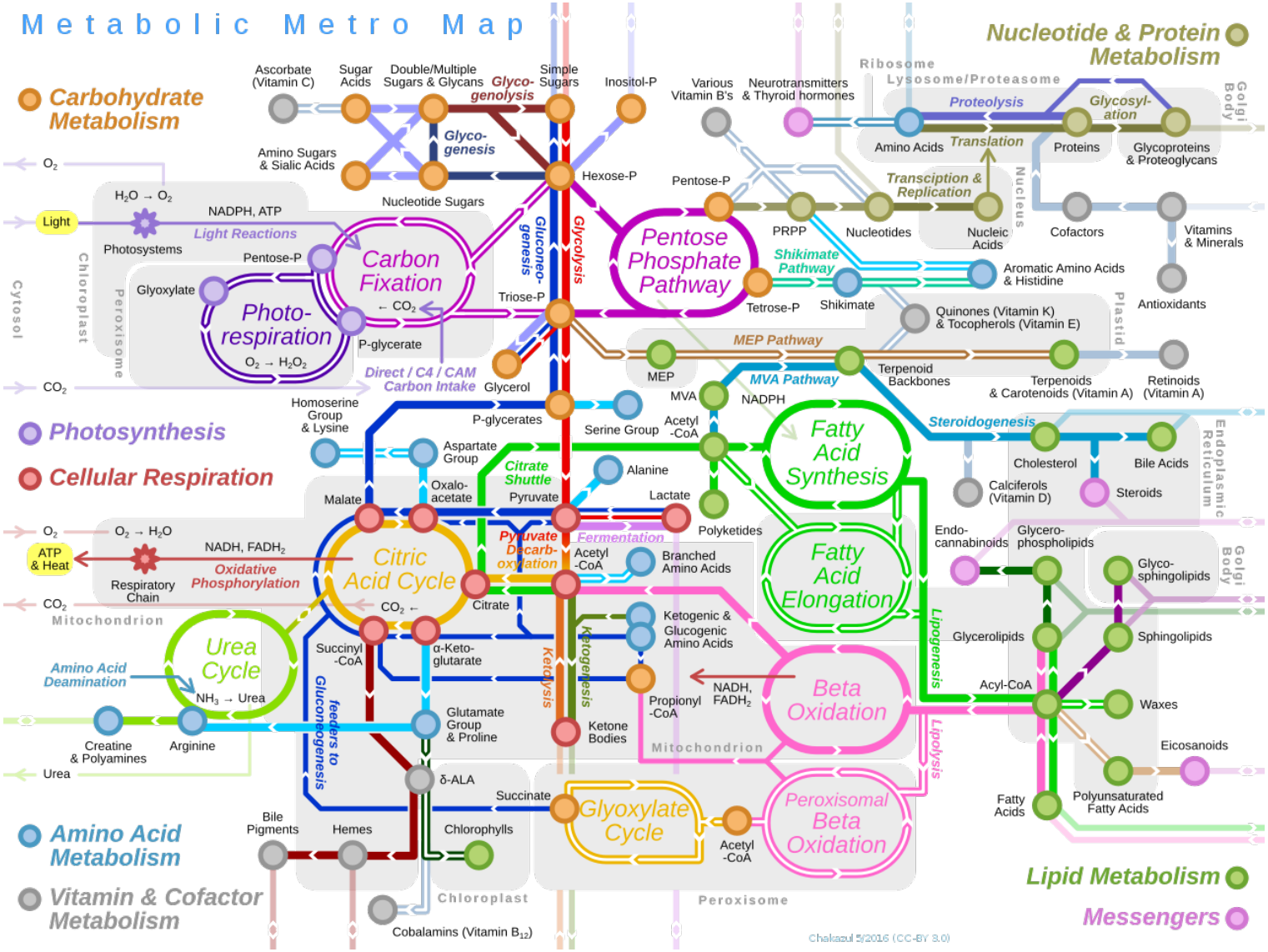
- Catabolic: break down components, e.g., glucose to pyruvate
- Anabolic: synthesize new components, e.g., carbohydrates, nucleic acids, ...
- Example: tricarboxylic acid (TCA) cycle



(from: https://en.wikipedia.org/wiki/Citric_acid_cycle)

Metabolic processes

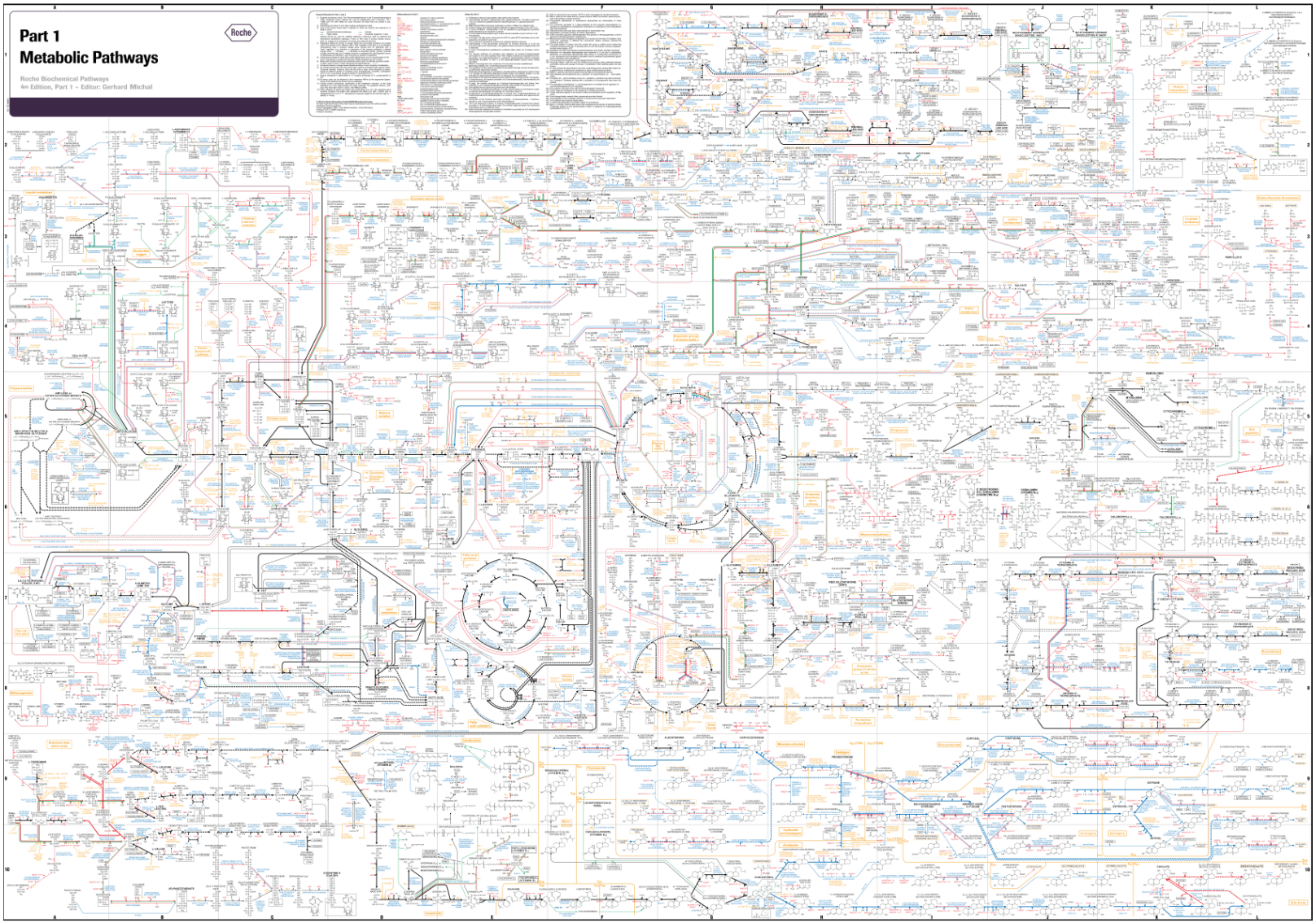
The cellular metabolism (simplified):



(from: `https://en.wikipedia.org/wiki/Metabolic_pathway`)

Metabolic processes

The cellular metabolism:



(Interactive version: `http://biochemical-pathways.com/#/map/1`)

How does the cellular system work?

The cellular system is **extremely** complex ...

If we want to know, how the system works, we need to know: *everything*

- How does gene transcription work?
  When are genes transcribed and at what level? (regulatory mechanisms)
  When are transcripts degraded?
- How does protein synthesis work? What quantities of proteins are produced?

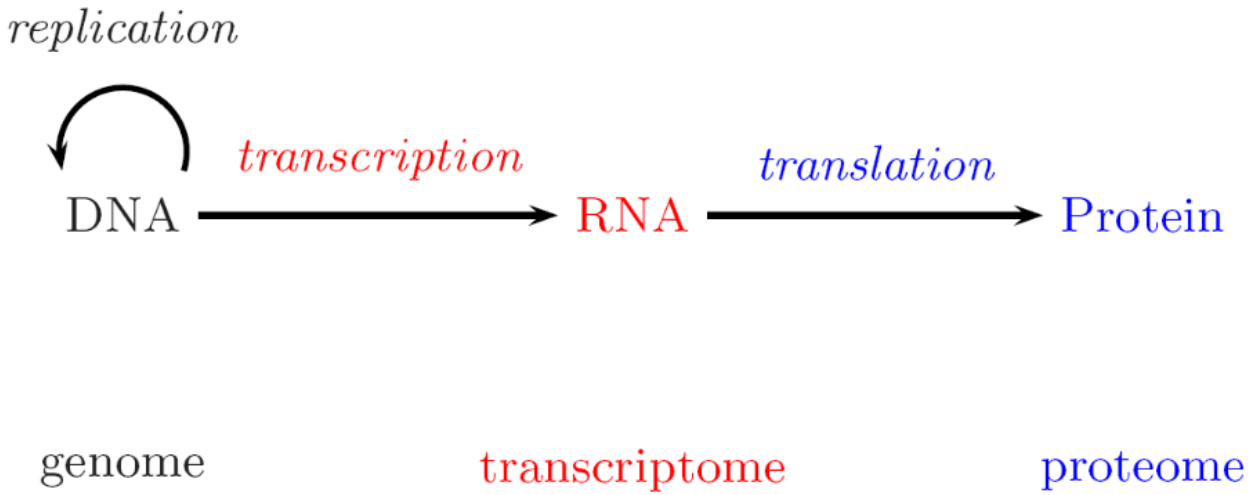
- How do specific molecules work (proteins, non-coding RNAs, ...)?
- How do specific molecules *interact* to exert a specific function?
- How are cellular processes intertwined and support or inhibit each other?

- How does the system *dynamically* change over time?
- How does the system change under certain conditions?

- How is it affected by disease (viruses, cancer, hereditary disorders, . . .)?
- . . .

“-omes” and “-omics”

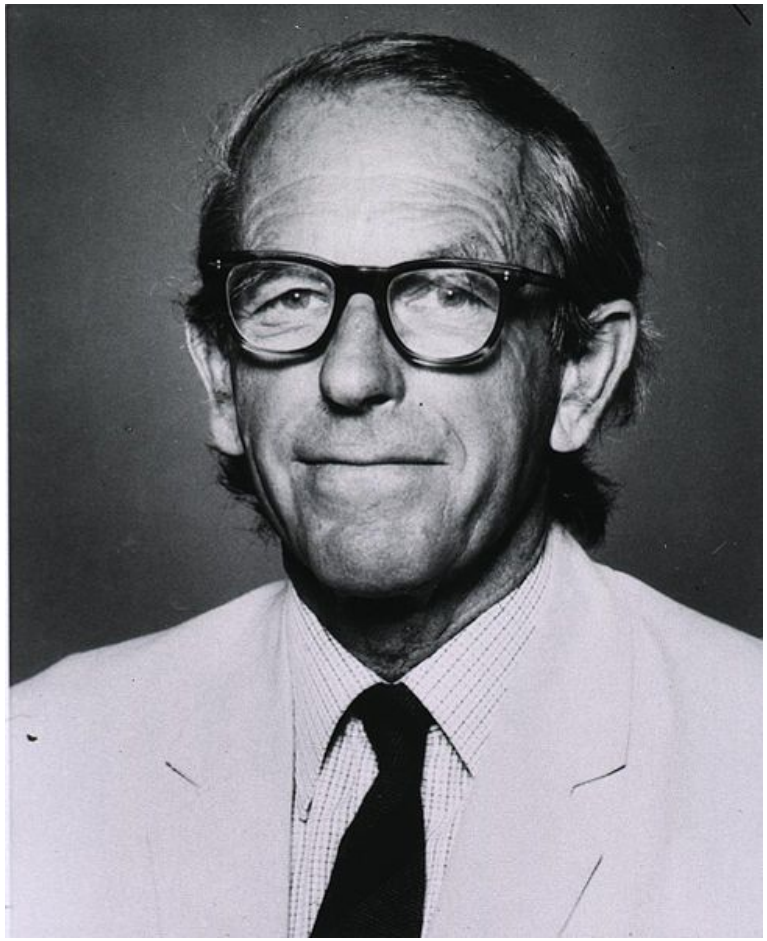
The Central Dogma revisited:



- “...ome”: the complete set of ...
  - genome: genes / genetic material
  - transcriptome: transcripts (state of the system in terms of gene expression)
  - proteome: proteins
  - interactome: molecular interactions
  - metabolome: chemicals involved in metabolic reactions
  - ...
- “...omics”: the study of ...
  - (see above)
  - ... by means of **high-throughput methods**!

Fred Sanger: $1\frac{1}{4}$ Nobel prizes

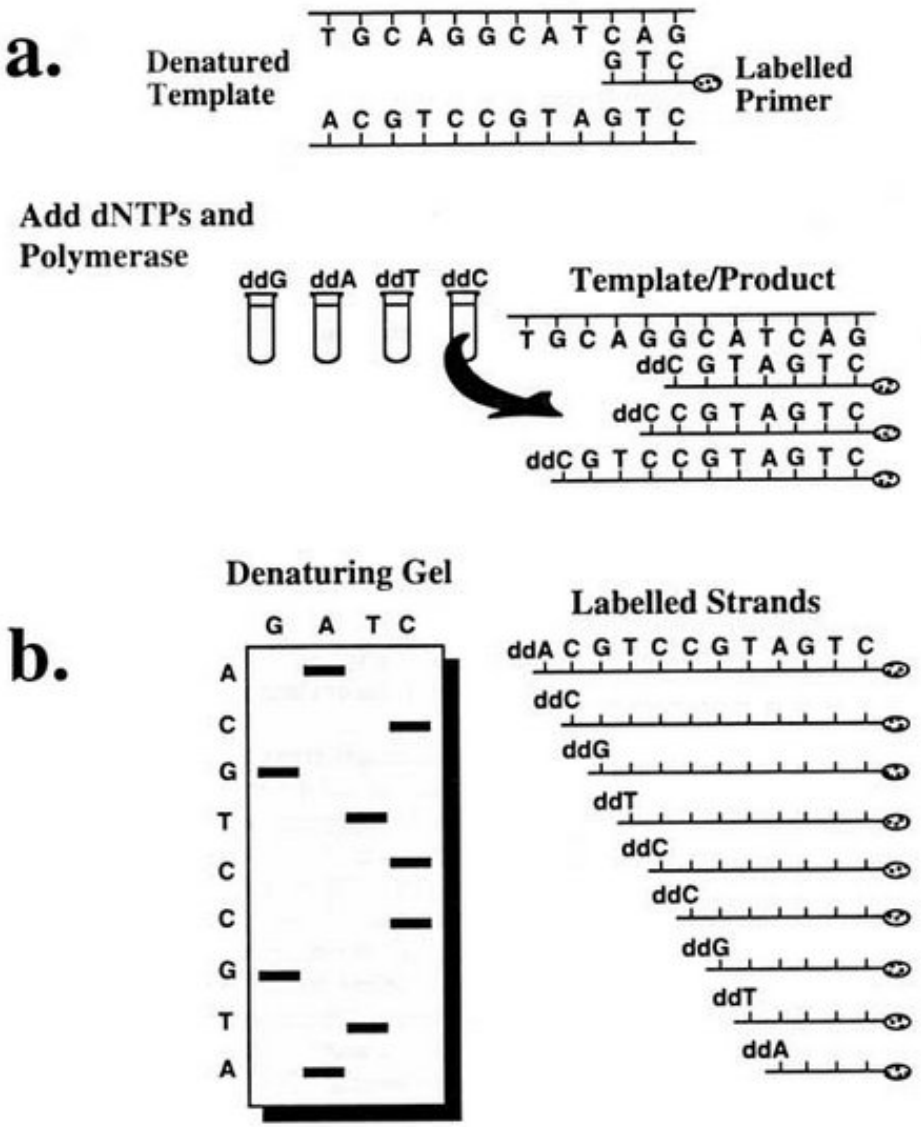
- 1958: Nobel prize in chemistry "for work on structure of proteins, especially that of insulin"
- 1980: Shared half of the Nobel prize in chemistry with Walter Gilbert "for determination of base sequences in nucleic acids"
- The major method of sequencing nucleic acids until recently
- Still used for confirming mutations from high-throuput sequencing data



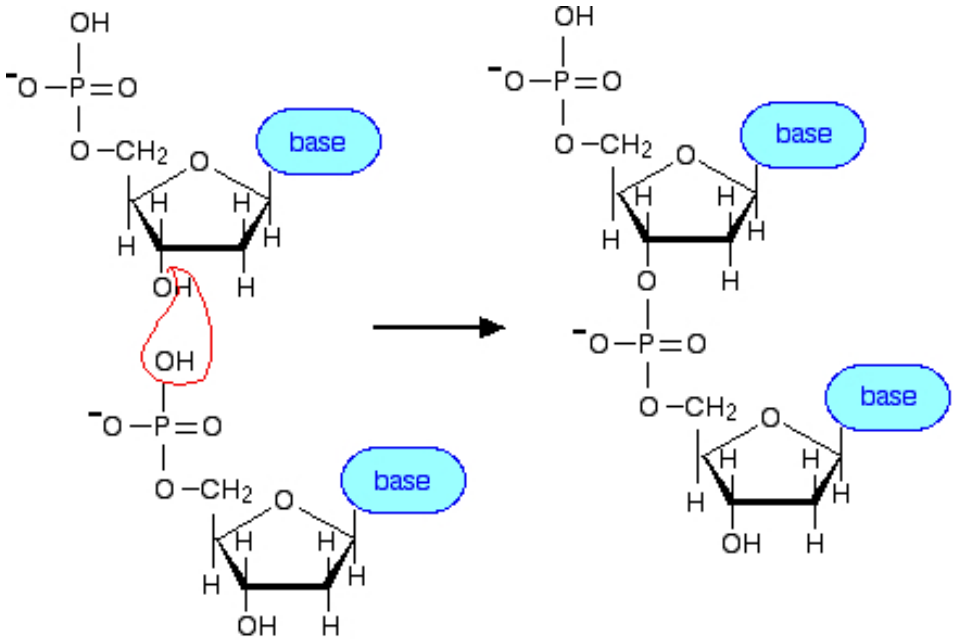
1918–2013
British biochemist

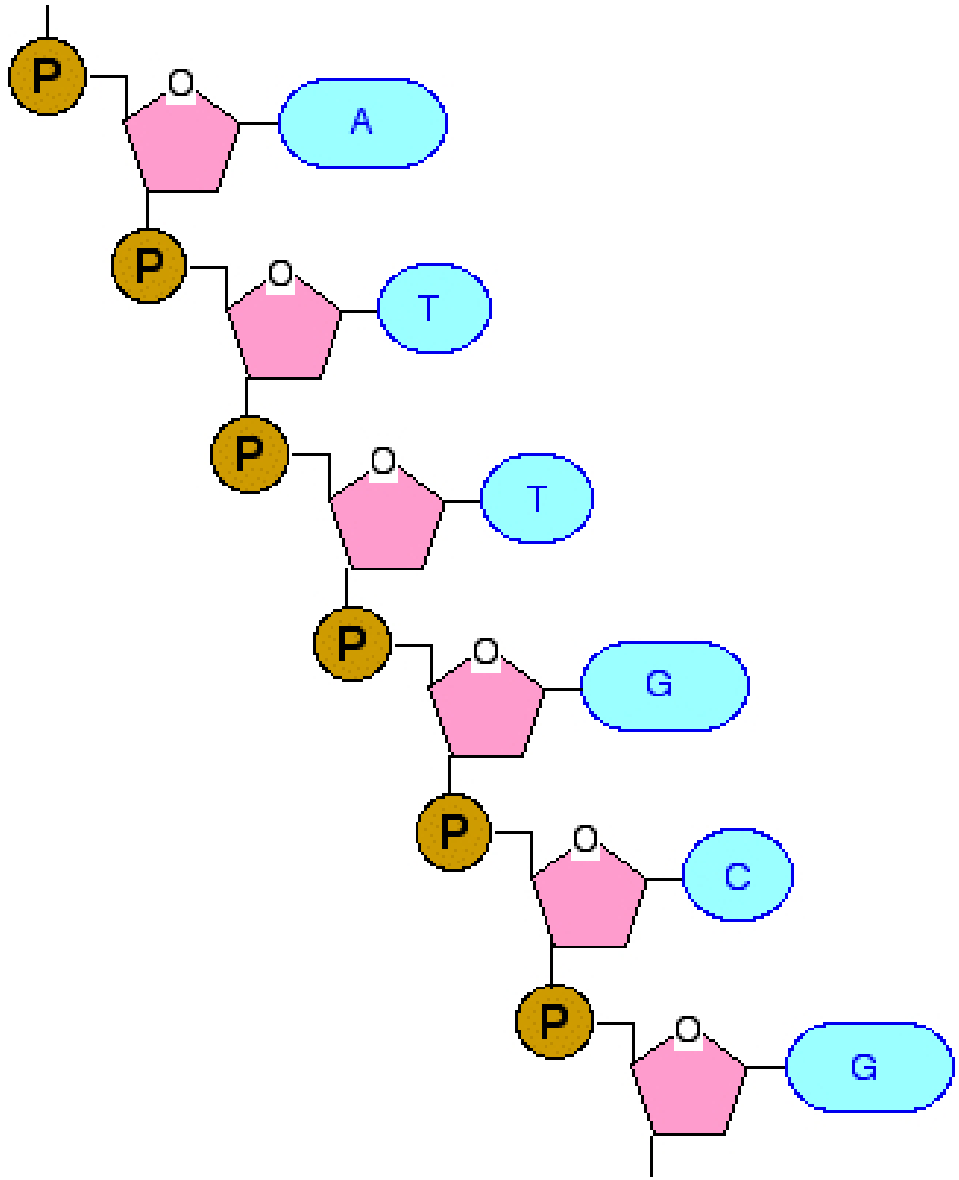
Sanger sequencing

- Sequencing by synthesis (of the complementary strand)
  - Basic idea: start from single-stranded DNA, reconstruct the reverse strand (similar to DNA replication) and check which bases you have to add to reconstruct it!
- Classical chain-termination method
- Ingredients:
  1. single-stranded DNA template
  2. DNA primer
  3. DNA polymerase
  4. normal deoxynucleotide-triphosphates (dNTPs)
  5. “chain-terminating”, dideoxy NTPs (ddNTPs)



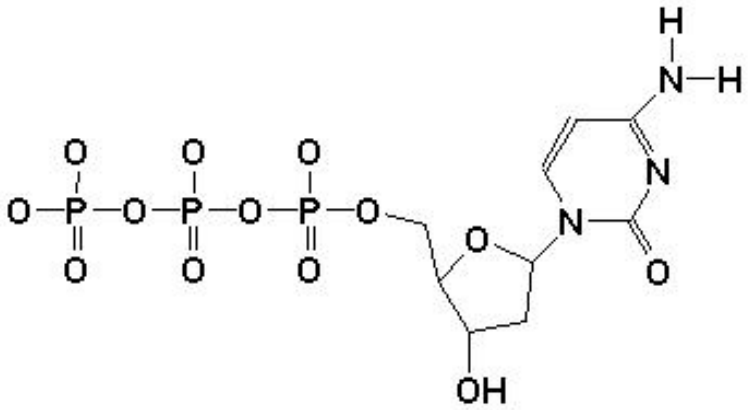
DNA synthesis: chain extension



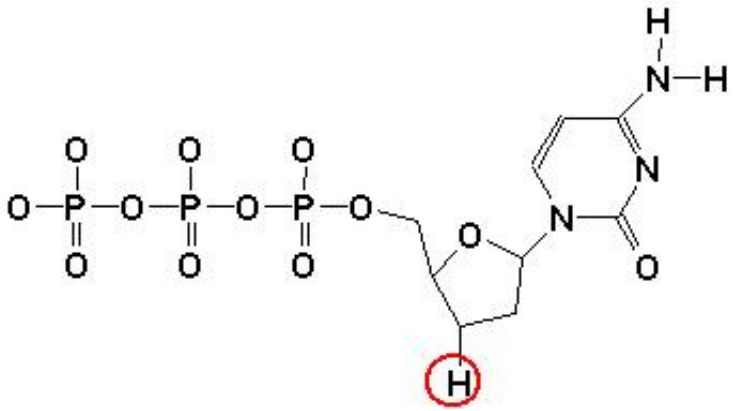


- Recall that DNA is extended from 5' to 3'

Key idea of Sanger sequencing: dideoxy nucleotides



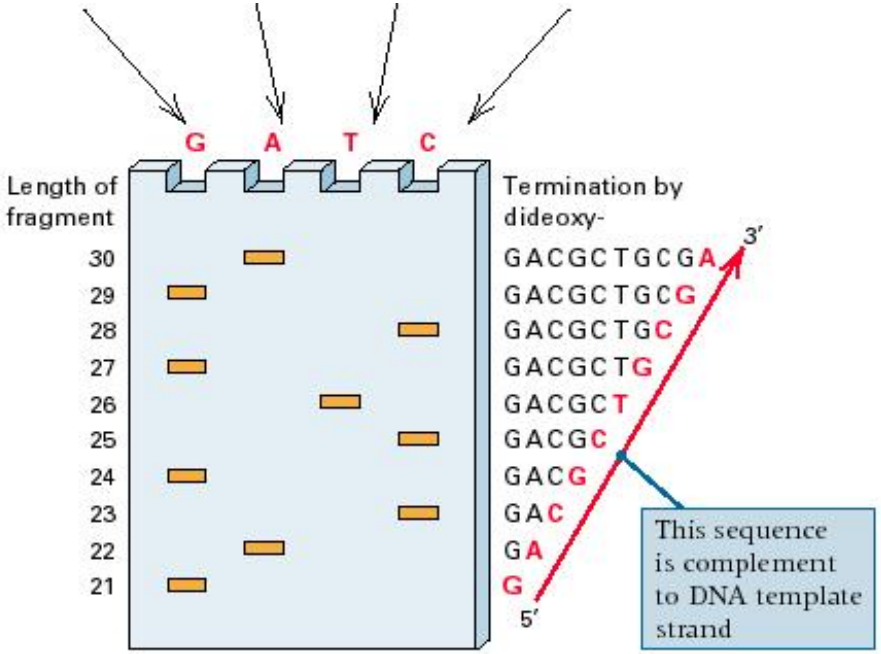
Example: deoxycytosine (dCTP)

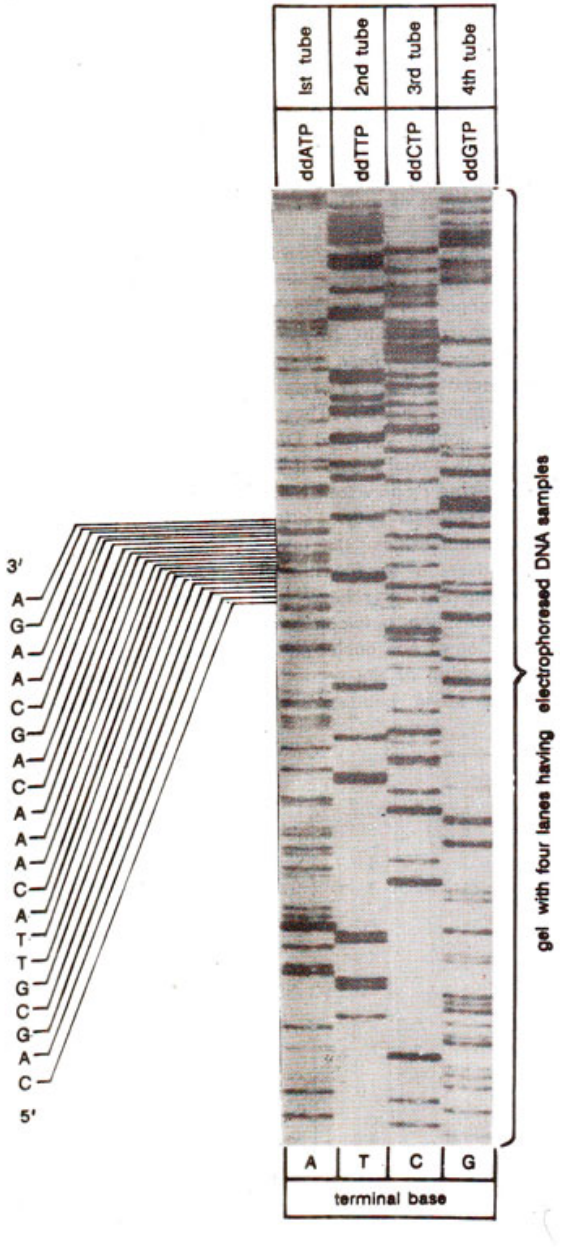


dideoxycytosine (ddCTP)

- A dideoxy nucleotide (ddNTP) stops DNA synthesis at specific nucleotides.
- For example, if the ddCTP to the right is incorporated into a growing strand of DNA, the lack of a free OH group at the $3'$ carbon would prevent the next nucleotide from being added and thereby terminate the chain extension!

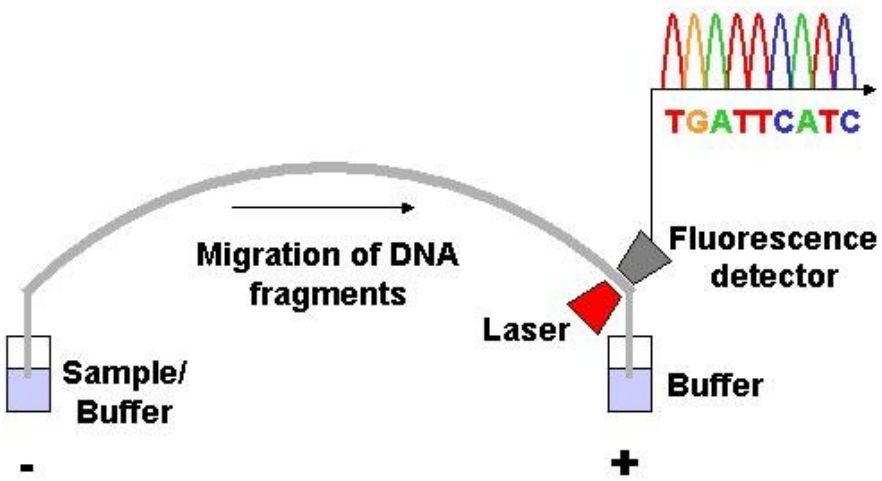
Sanger sequencing: radioactive

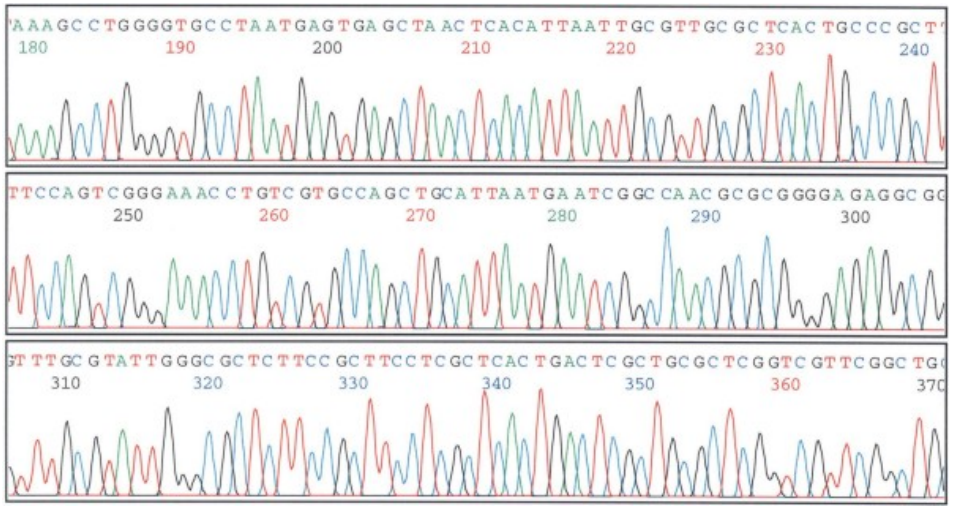




- Initially done with radioactively marked terminating nucleotides, e.g., ddATP–[α-^{33}P]
- Labeled primers, four reactions, each with a different ddNTP
- Electrophoresis: use an electric field to attract molecules across a gel; larger molecules (here: longer sequences!) will be slower!

Sanger sequencing: fluorescent





- Dye-terminator sequencing
- Label each of the 4 ddNTPs with different fluorescent dye (different wavelength)
- Conventional colors: A, T, C, G
- Allows one reaction, rather than four reactions as in the labeled-primer method.
- Plot intensity of each wavelength against electrophoresis time (“chromatogram”)

Sanger sequencing: fluorescent

- Sanger sequencing powered the characterization of the human genome
- Very accurate determination of a sequence

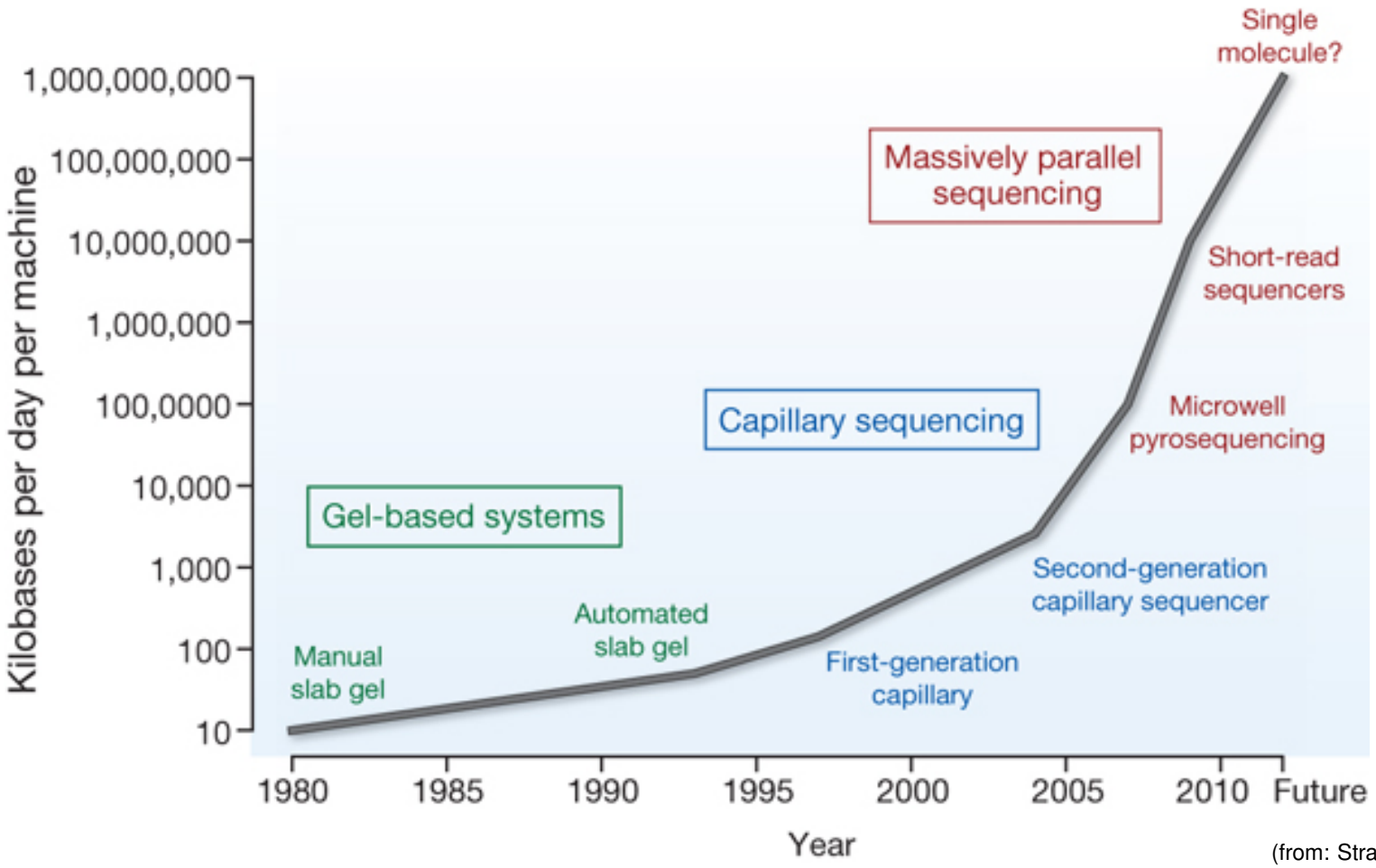
- But: still extremely limited in throughput
- At height of Sanger era, 400 kb (kilobases) per machine per day
- about 45 thousand runs needed for one human genome at 6-fold coverage if everything works perfectly...





International Human Genome Sequencing Consortium (Lander et al., 2001): Initial sequencing and analysis of the human genome, *Nature* **409**:860–921

Next-generation sequencing (NGS)

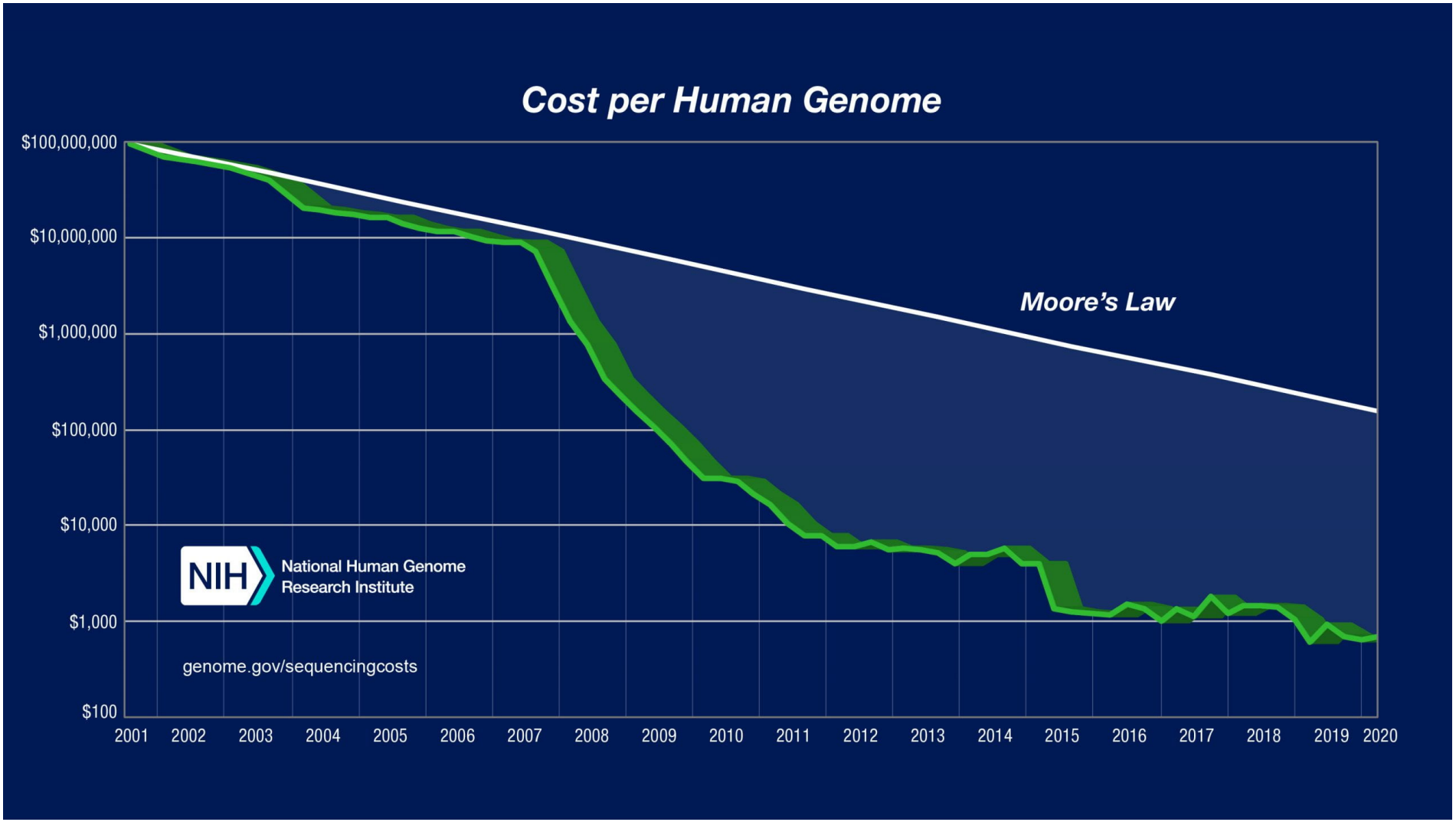


(from: Stratton et al., 2009)

- NGS: refers to one of a number of technologies that enable a massive parallelization of DNA sequencing

Next-generation sequencing (NGS)

Due to fast technological advances, the cost of whole-genome sequencing (WGS) has dropped significantly!



Implication: we're producing **huge** amounts of sequencing data!

Next-generation sequencing (NGS)





- NCBI GenBank: Oct. 2005 about 50 Gb (gigabases) sequence data
- Illumina GA at 1000 Genomes project in year 2008 – 2,500 Gb
- “Each week in Sept–Oct of 2008, the 1000 Genomes Project created the equivalent of all the data in GenBank”
- 100,000 Genomes project in the UK (100K+ genomes in december 2018)
- Currently in the EU: 1+ Million Genomes (1+MG) initiative and Beyond 1 Million Genomes (B1MG) project

Illumina sequencing

- There have been several competing NGS technologies
- At present, the technology of Illumina seems to be superior for most applications
  - At least for “next-generation” sequencing; for “third-generation” sequencing, or long-read sequencing, other technologoies are used

- Four Basic Steps:

1. DNA & library[1] preparation	Make random DNA fragments, append adapters
2. Chip/flowcell prep	Attach fragments to surface and amplify
3. Sequence	Massively parallel DNA sequencing
4. Analyze	Bioinformatics starts here ...

- Understanding 1.–3. is essential for 4.! [but not relevant for our exam ...]
  - To be able to process error-prone biological data, it is important to know how such data is experimentally produced!
  - (But for the algorithms we will see in this course, this is less important; we’ll mostly assume the data is error-free ...)

[1] Biological/molecular library, not programming library!

Library prep

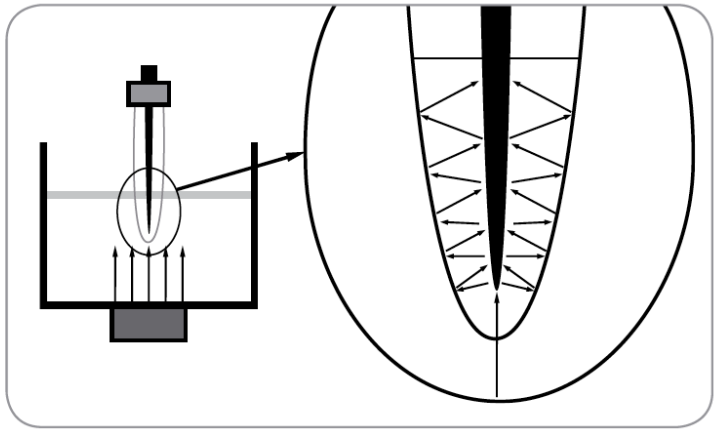
Purpose of library prep: create collection ("library") of *random* DNA fragments ready to be sequenced!

- Four major substeps for library prep (about 6 hours of lab work)
  1. Fragment the genomic DNA (after having extracted it from cells)
  2. Repair ends and add an A overhang
  3. Ligate adapters to DNA fragments
  4. Select ligated DNA (not all fragments will have adapters after this procedure; we need to remove those who don't)

Library prep (1): fragmentation

- Most Illumina protocols require that DNA is fragmented to less than 800 nt
- Ideally, fragments should have uniform size (in reality: fragment size follows a hopefully narrow distribution)

- Fragmentation, for example, by
  - Sonication: uses ultrasound waves in solution to shear DNA
  - Nebulization: forcing solution of genomic DNA to pass as a fine mist through a small hole in the nebulizer; repeatedly done



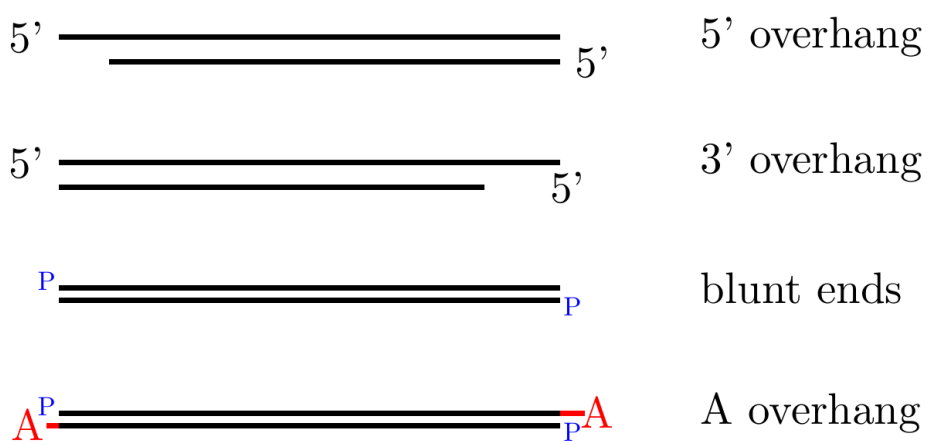


(from: Bioruptor, `http://www.diagenode.com/` (left); AIR™ DNA Fragmentation Kit, `http://www.biooscientific.com/` (right))

Library prep (2): end repair

Purpose of end repair: downstream enzymatic steps won't work unless the DNA fragments are *nice*!

- After fragmentation, DNA fragments can have $5'$- or $3'$-overhangs which need to be repaired:
  - Reconstruct reverse strand at $5'$ overhangs (DNA polymerase)
  - Cleave away $3'$ overhangs (exonuclease)
- Need also to be sure that the phosphate (P) group isn't missing at the $5'$ ends
- Moreover: A-tails (single adenine) are added for downstream steps
  - Purpose: keep DNA fragments from ligating with each other & enable ligation to adaptors (which cleverly are designed to have a 'T'-base overhang)



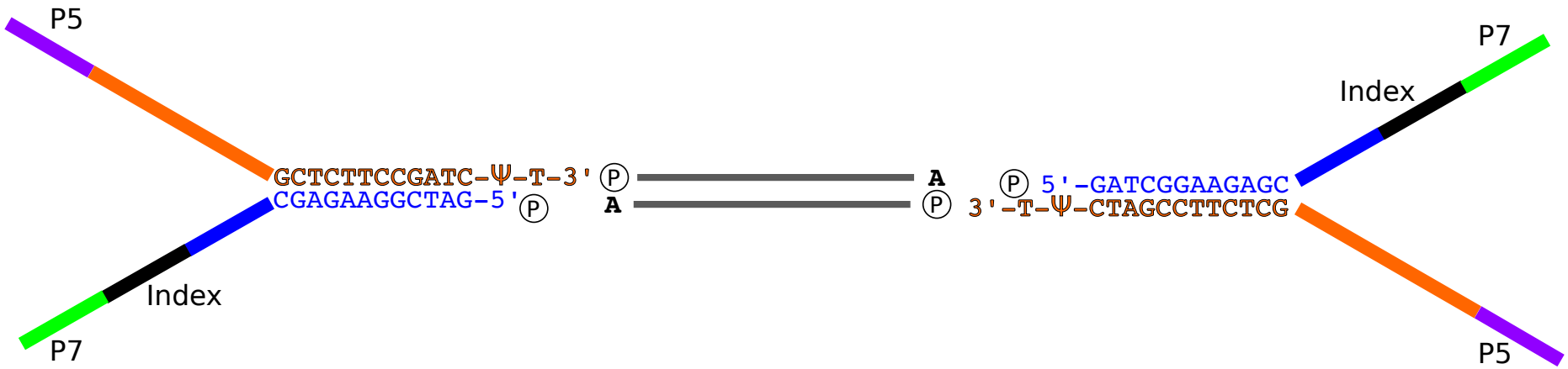
Library prep (3): adapter ligation

Purpose of adapter ligation: adaptors get attached (ligated) to the end-repaired DNA fragments. This enables three things:

1. The adaptors will later bind to complementary oligonucleotides on the flowcell
2. Allow PCR enrichment (amplification) of adapter-ligated DNA fragments only; not shown here ... (PCR = polymerase chain reaction)
3. Allow for indexing or "barcoding" of samples (barcode shown as NNNNNN below)

TruSeq Universal Adapter (P5)
5 AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT 3

TruSeq Indexed Adapter (P7)
5 GATCGGAAGAGCACACGTCTGAACTCCAGTCAC–NNNNNN–ATCTCGTATGCCGTCTTCTGCTTG 3



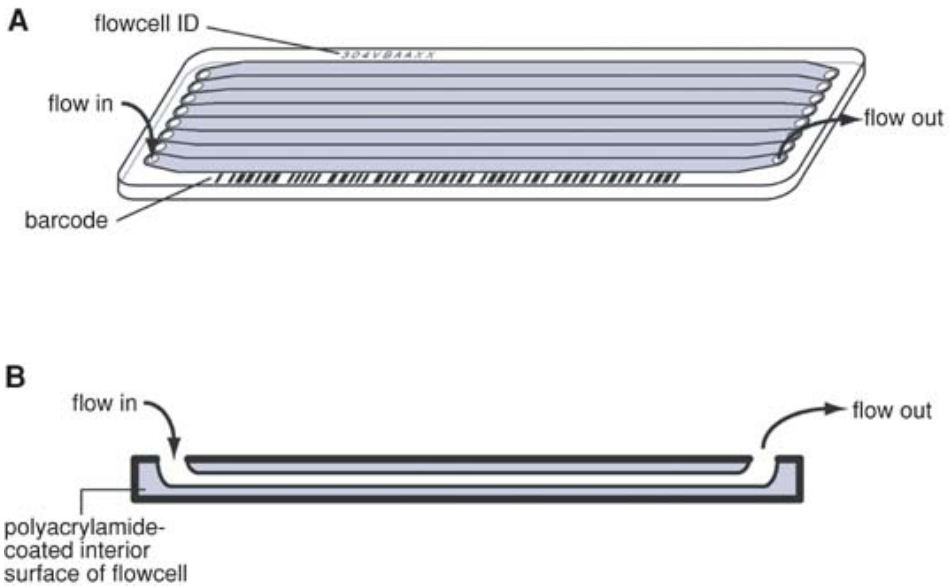
Flow-cell prep

Purpose of flowcell prep:

- Bind ligated DNA fragments to flow cell
- Perform in site amplification of individual molecules (creating clusters or "colonies") to boost signal strength later during sequencing

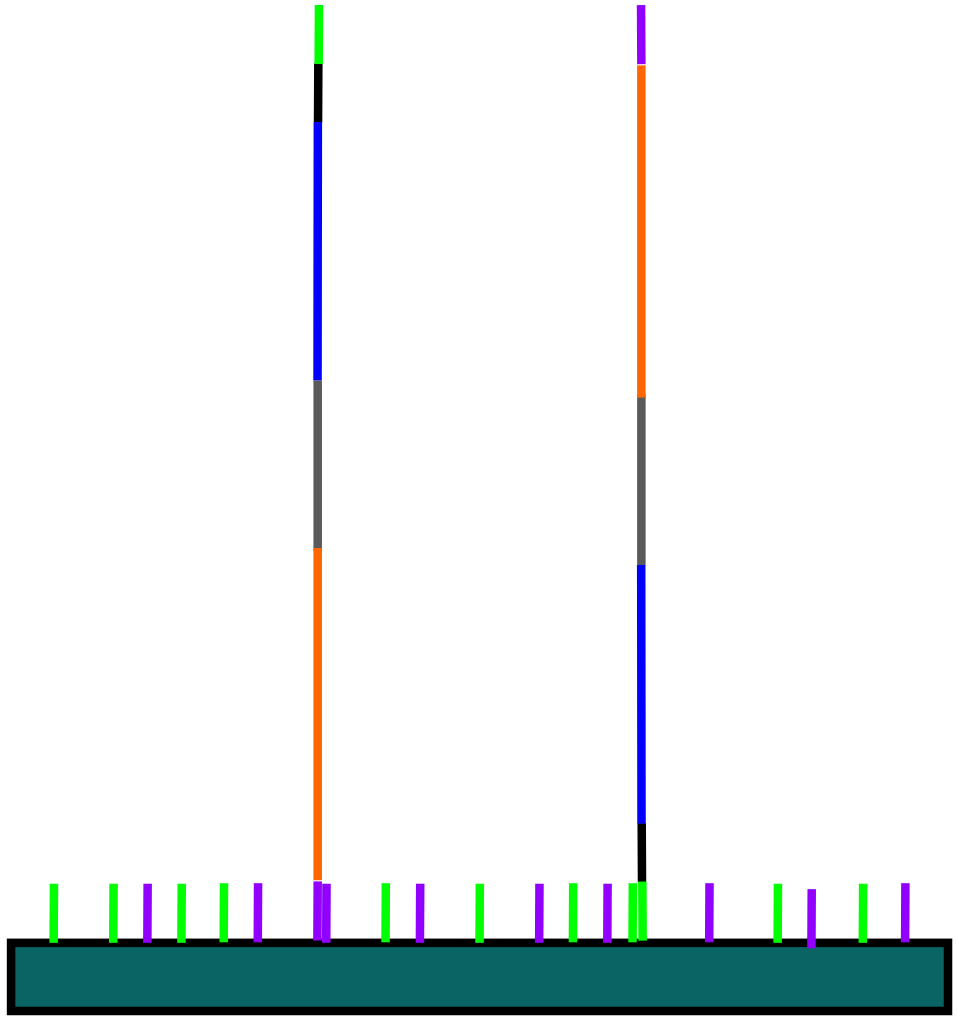
Illumina flow cell: hollow glass slide with eight channels ("lanes")





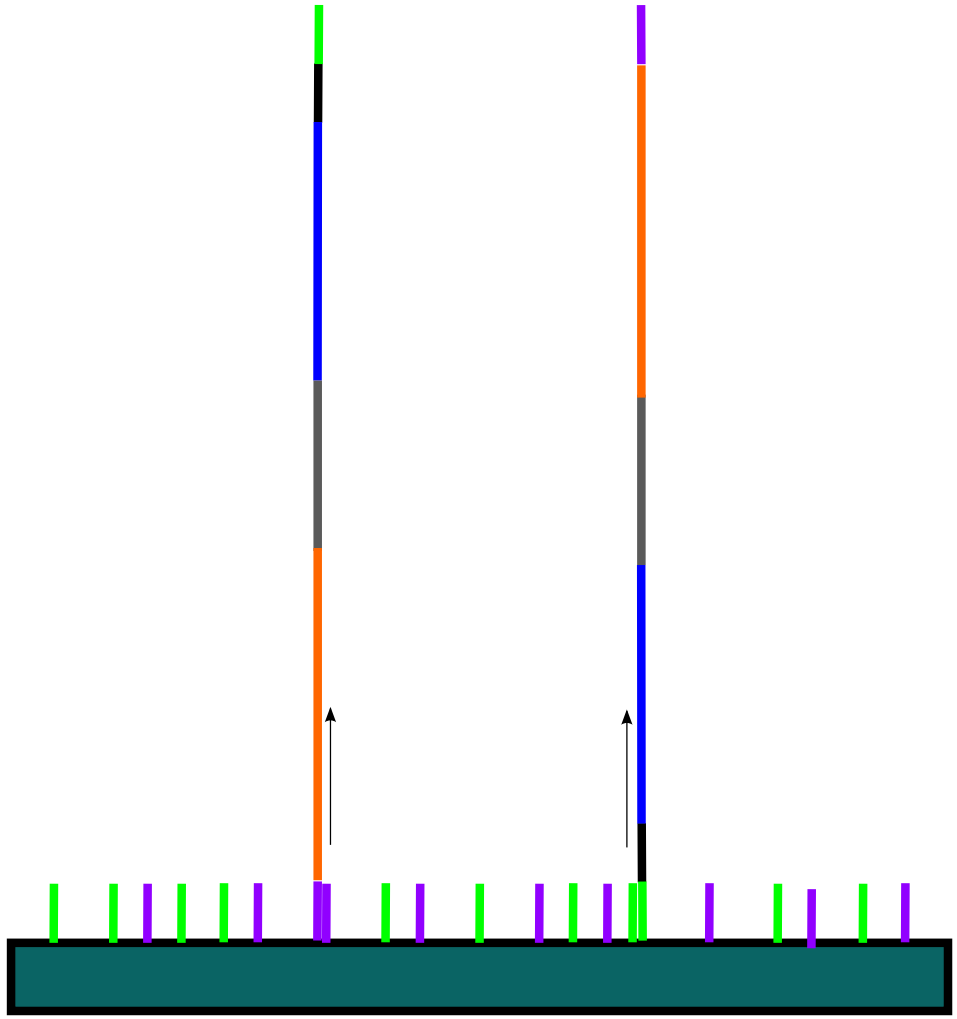
Flow-cell prep (1): library deposition

- The double-stranded adapter-ligated DNA fragments are denatured with NaOH
- The single stranded DNA molecules are then transfered to the flowcell
- Here, they bind to oligonucleotides (“oligos”) attached to the flow cell surface; these oligos are designed to be reverse complementary to sequences in the adapters



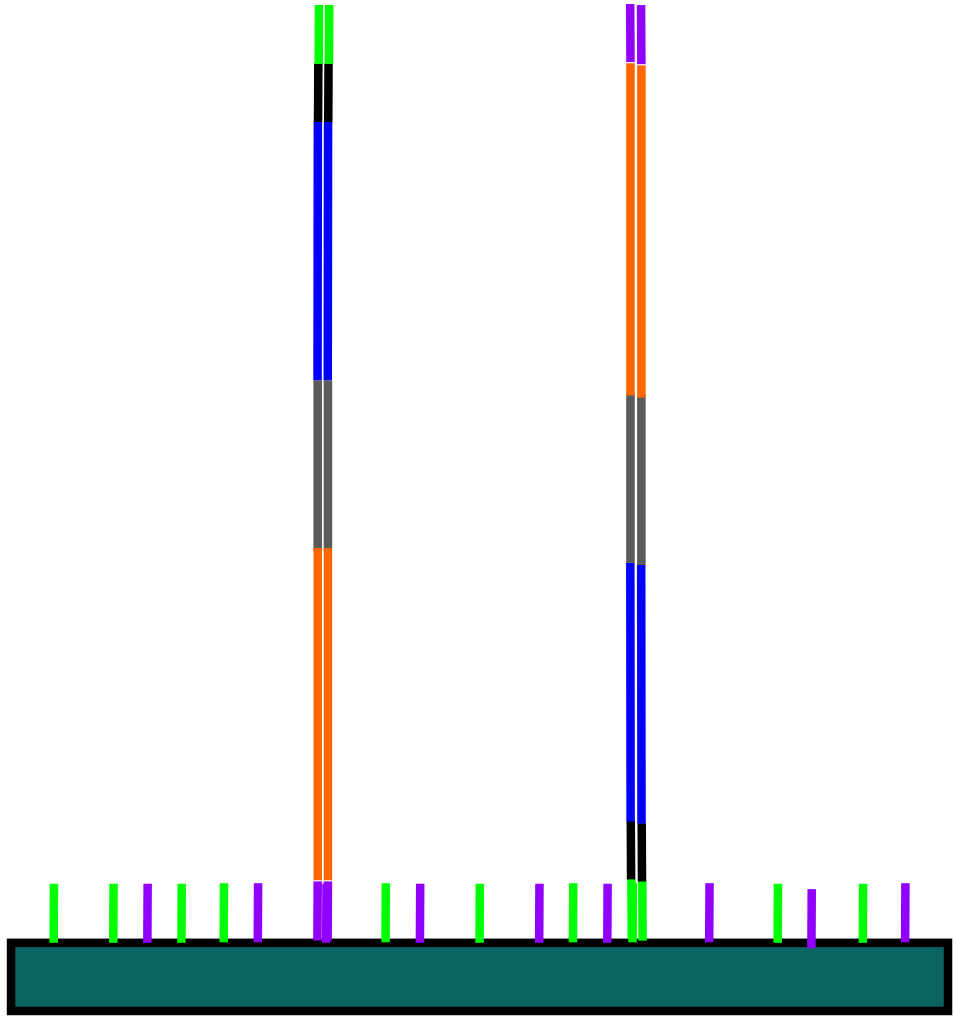
Flow-cell prep (1): library deposition

- Extension mix (buffer, dNTP's, Taq polymerase) is pumped into a channel
- The oligos on the flowcell are elongated using the ligated DNA fragments as a templates (reconstruction of the reverse strand of the DNA fragments to be sequenced)



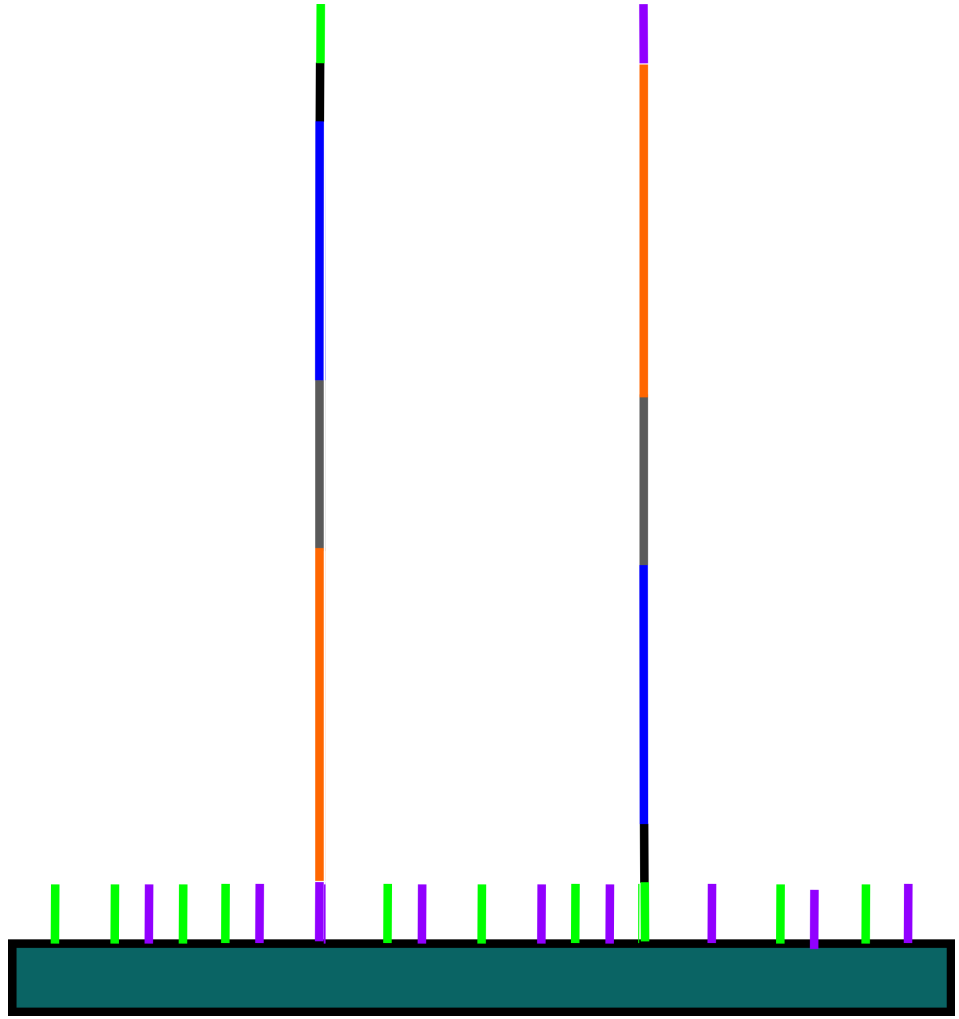
Flow-cell prep (1): library deposition

- We now get a double-stranded molecule



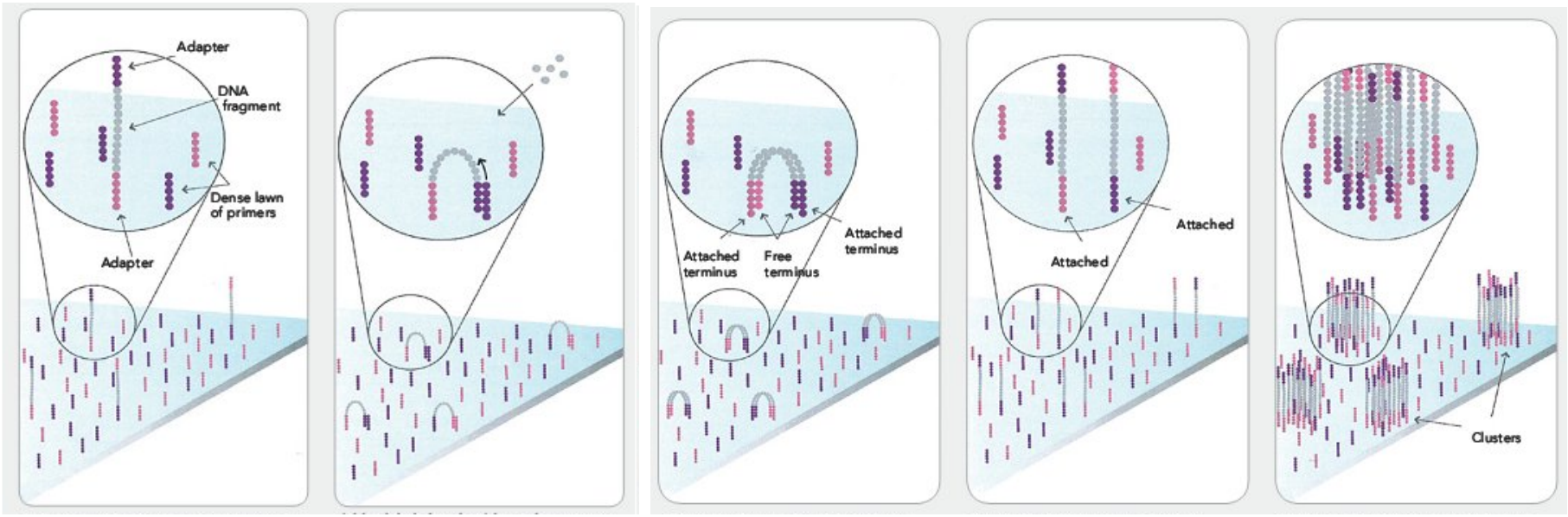
Flow-cell prep (1): library deposition

- Formamide is added to denature these products, and the original fragment (which is not covalently attached to the flowcell) is washed away
- The reconstructed reverse complement remains covalently fixed to the flowcell!



Flow-cell prep (2): bridge amplification

Purpose: amplify single molecules attached to flowcell as described above by factor of $\sim$ 1000 to generate more signal in the following sequencing steps



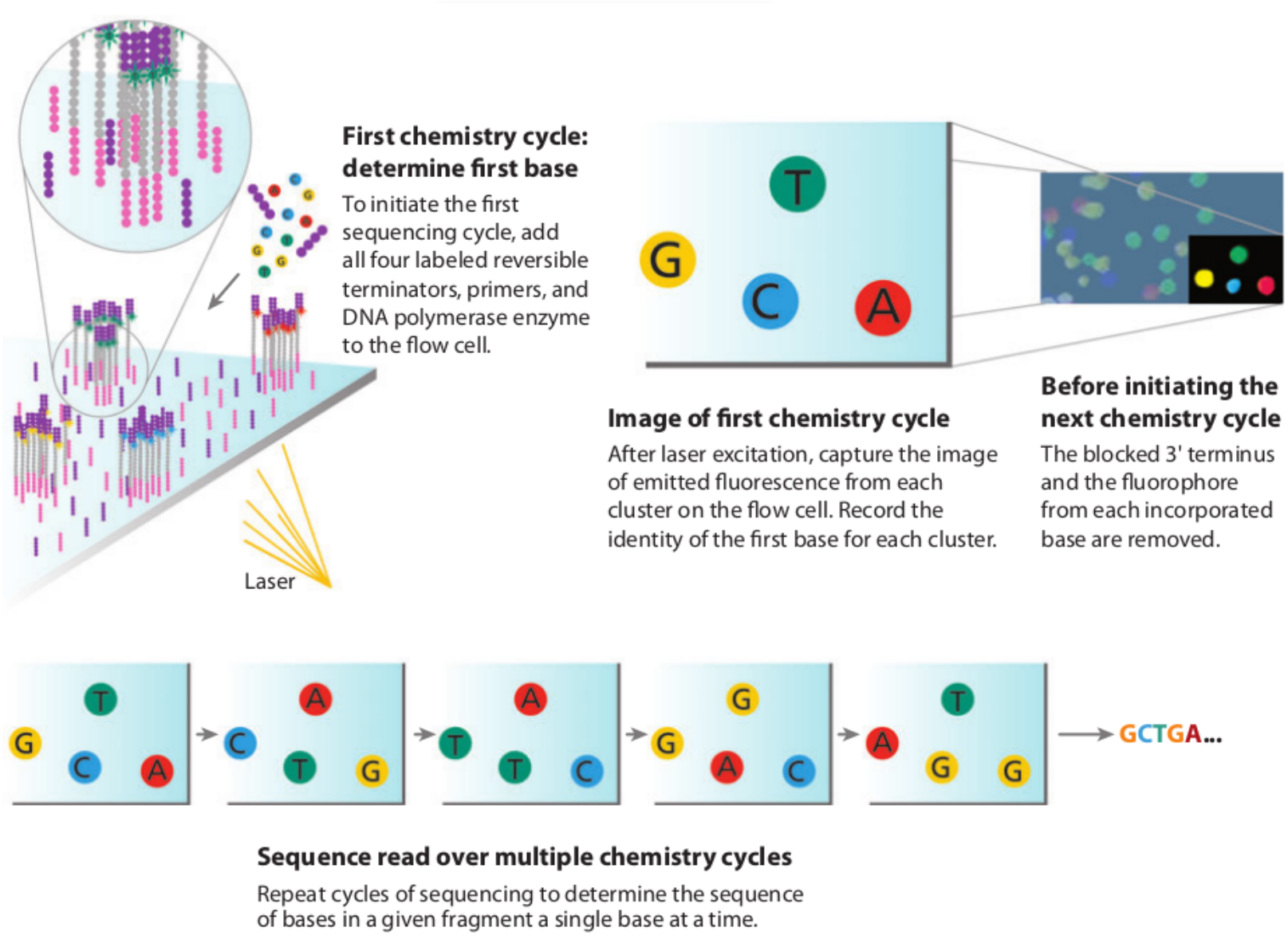
- Single-stranded DNA molecule (1) bends and hybridizes to a second oligo on the flowcell surface (2)
- Extension (reconstruction of reverse complementary strand) is performed all the way back to the other oligo: a “bridge” of double-stranded DNA (3)
- Each strand of the bridge is covalently attached to a different oligo! (3)
- Denaturation seperates the two strands, so that they again can bend and hybridize to another oligo (repeat from step 1, now with twice as many fragments!)
- This is repeated several times to create a cluster of copies of the original DNA fragment!
- Finally: both strands are present; keep only fragments from one of the two strands, those of the reverse strand are cleaved away

Sequencing by synthesis

- Sequencing by synthesis (SBS) is relatively simple to understand after all of the prep steps

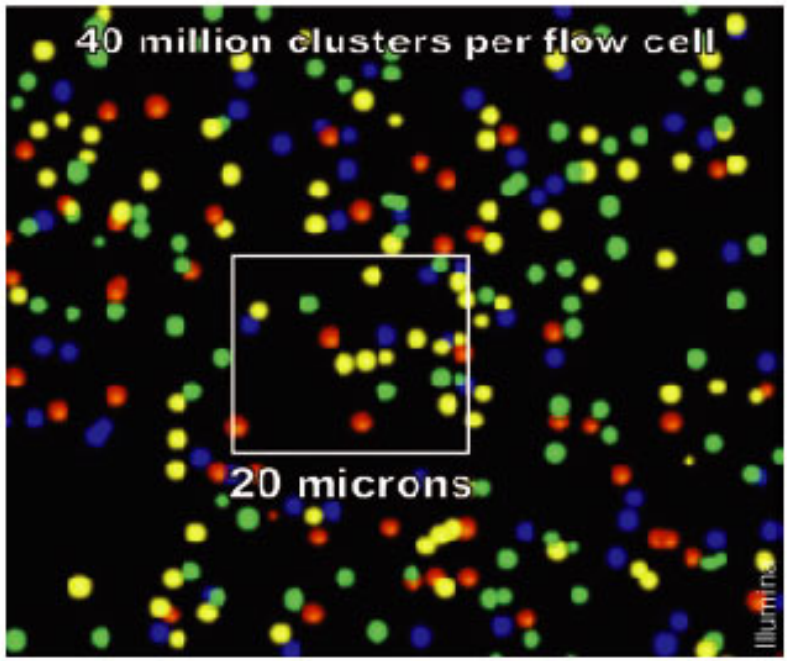
- Basic idea: reconstruct the reverse strand, but in each "cycle" add **only one** nucleotide and identify it (A, C, G or T?)
  1. Incorporate one base at a time using four differently marked ddNTPs (fluorescent dyes of different colors)
     - ⋆ Difference to Sanger sequence: the ddNTPs have *reversible* terminators (reversible block of the $3'$ OH group)
  2. Contemporaneously identify which base was incorporated in each DNA fragment cluster by measuring the wavelength of the incorporated ddNTP
  3. Unblock the ddNTP (remove the terminator such that the sequence can be extended in the next cycle) and repeat!

Sequencing by synthesis



Mardis E (2008) *Annu Rev Genomics Hum Genet* **9**:387–402

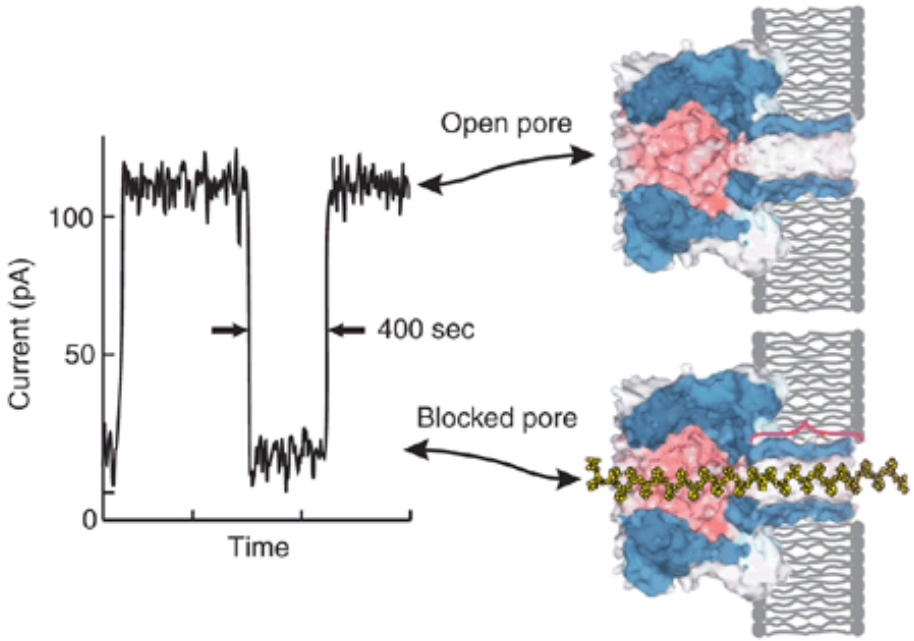
Sequencing by synthesis



- An imaging step follows each base incorporation step (we speak of one “cycle”)
- After each imaging step, the $3'$ blocking group is chemically removed and the cycle is repeated
- After having completed all cycles:
  - A “base-calling” algorithm assigns sequences and associated quality values (“base qualities”) to each “read”
  - Read = the sequence of bases called from the colored images for one specific spot of the flow cell, corresponding to one specific DNA fragment!
  - A quality checking pipeline removes poor-quality sequences

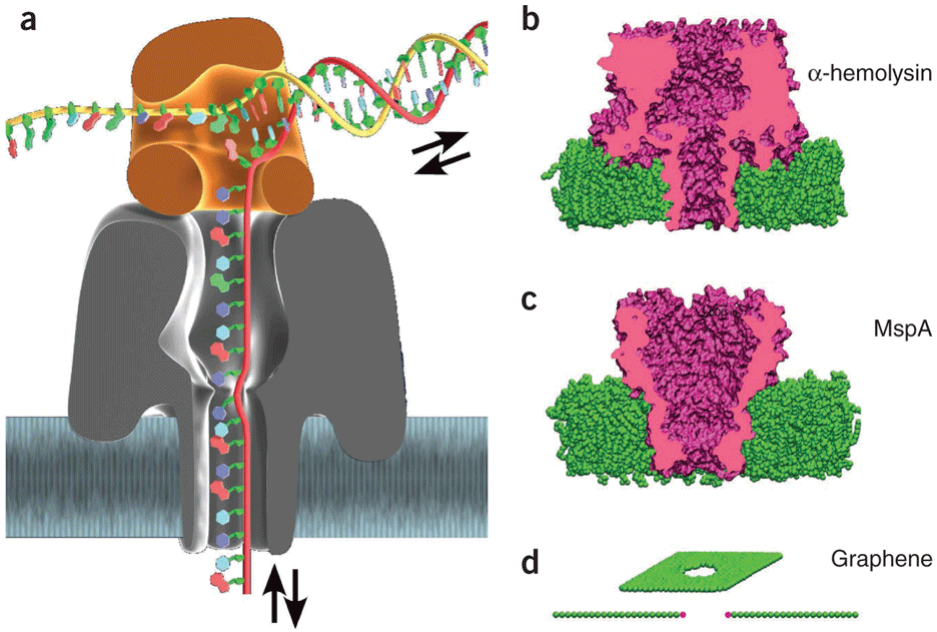
Nanopore sequencing

- A nanopore: A hole of $\sim$ 1 nm in diameter
- Apply voltage across pore and observe current due to conduction of ions through the nanopore
- The different bases of DNA block the hole to different extents and change the current in a characteristic way



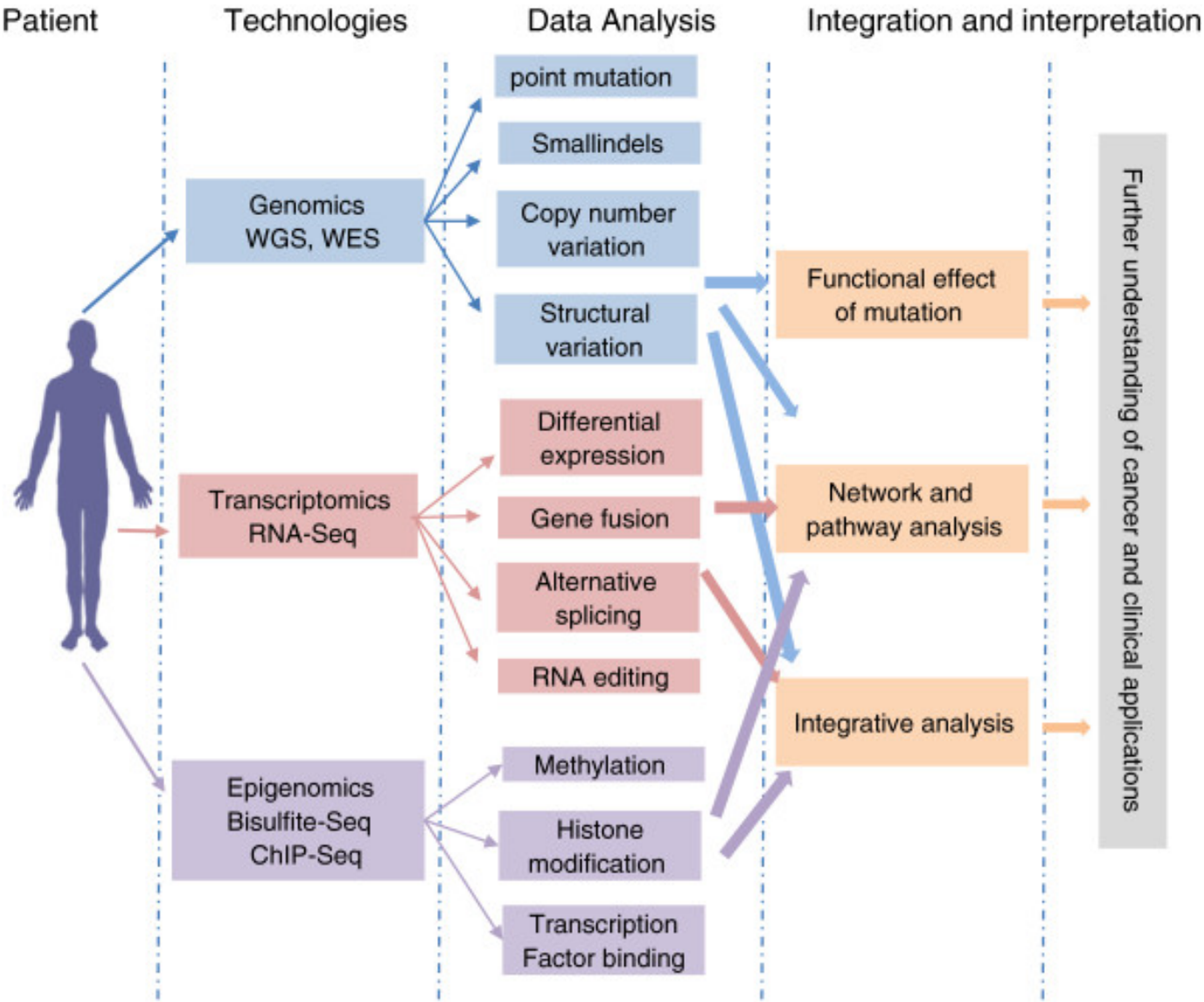
Nanopore sequencing

- By following changes of current over time, we can determine the sequence
- True single-molecule sequencing!
- Advantages: 1) Long reads ($>$10,000 nt); 2) cheap (!)
- Major disadvantage: higher error rates due to signal-to-noise issues inherent in the technology



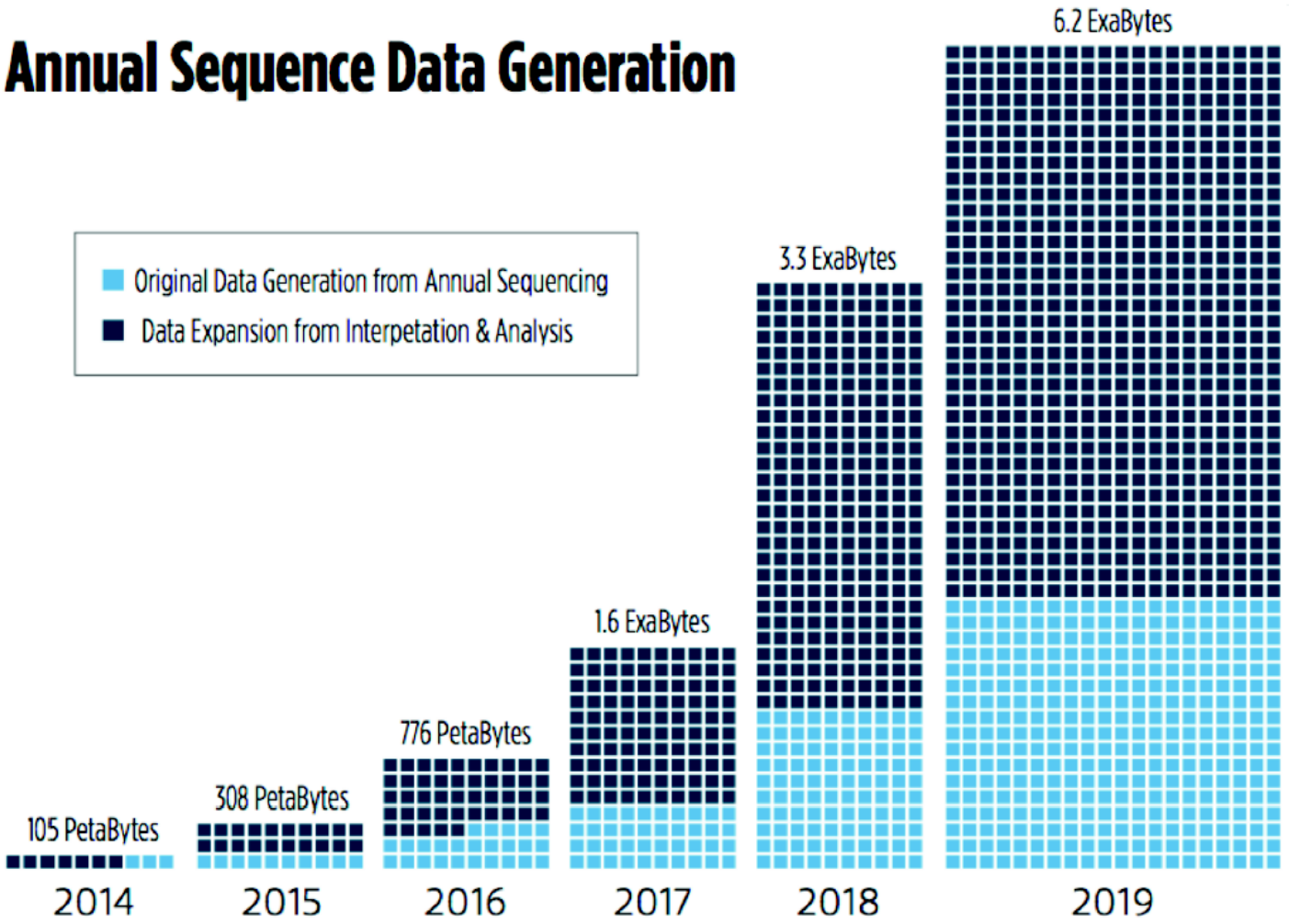
Do we sequence only the genome?

Beyond DNA-seq:

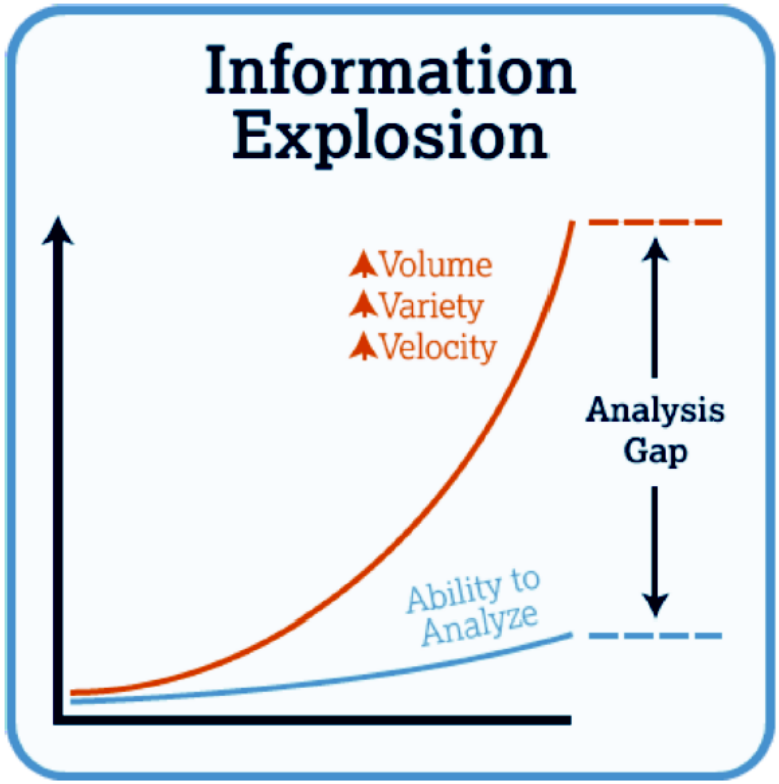


(from: Shyr and Liu, 2013)

Data: only sequences?



Data analysis?



"Biology easily has 500 years of exciting problems to work on."
Donald Knuth

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Some introductory texts and text books I



Bruce Alberts et al. (to be released in July 2022)
Molecular Biology for the Cell (7th edition)
Publisher: W.W. Norton & Company (previously: Garland Science)



Bruce Alberts et al. (to be released in May 2022)
Essential Cell Biology (5th edition)
Publisher: W.W. Norton & Company (previously: Garland Science)



Lawrence Hunter
Molecular Biology for Computer Scientists
https://tandy.cs.illinois.edu/Hunter_MolecularBiology.pdf

Some videos I



MITx Bio – The Structure of DNA

https://www.youtube.com/watch?v=o_-6JXLYS-k



yourgenome – From DNA to protein

https://www.youtube.com/watch?v=gG7uCskUOrA



yourgenome – DNA replication

https://www.youtube.com/watch?v=TNKWgcFPHqw